



PE CIVIL-STRUCTURAL EXAM 03

Table of Contents

Problems.....	2
Solutions.....	39

Problems

[Click here for Solutions](#)

1. A floor system is constructed using a 2-inch steel deck and a 3-inch lightweight concrete topping. W8X18 Secondary Beams are spaced at 6 feet center to center. What is the factored dead load (to be used in combination with live loads) acting on each secondary beam?
Weight of steel deck = 5 psf

(A) 192 lb/ft
(B) 220.5 lb/ft
(C) 264.6 lb/ft
(D) 202.5 lb/ft

2. According to IBC 2018, which of the following factors does not affect the reduction of design live loads?

(A) The expected intensity of loading
(B) Tributary Area of the member
(C) The expected service life of the structure
(D) The utility of the structure

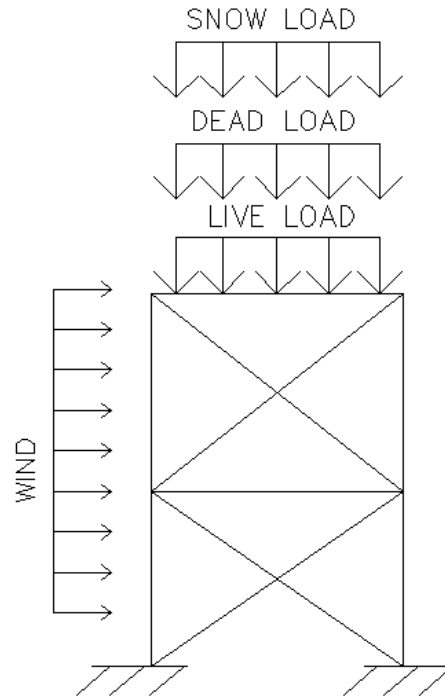
3. A school classroom has an area of 12ftX12ft. Furniture and equipment are estimated to weigh a total 2000 pounds. The class can seat only 10 students and 2 teachers for any class. What is the minimum total live load to be considered to design the classroom? Assume the maximum weight of each student and teacher to be 200 pounds.

(A) 4400 lb
(B) 5760 lb
(C) 7040 lb
(D) 14400 lb

4. Which of the following is not true about construction loads?

- (A) Construction loads can be higher than permanent live loads acting on the building
- (B) When considering construction loads, wind loads can be reduced, adjusting it based on the duration of construction.
- (C) The load factors for permanent live loads and construction loads are different in ASD methodology.
- (D) Concentrated loads are both static and dynamic.

5. Structural stability is to be evaluated for the framed structure below. The structure is located in a mountainous region with significant snow and wind load. To evaluate the stability of the structure against foundation uplift, which combination must be used?



- (A) Dead + Live + Snow + Wind
- (B) Wind + Snow
- (C) Dead + Wind
- (D) Live + Wind

6. An HVAC unit is to be installed at the roof of a 5 story building. The unit weighs 200 pounds. The unit is 6 ft tall. The length and width of the unit is 5ft each. Center of gravity of the unit can be assumed to be at the geometric center of the unit. Calculate the lateral wind load acting on the unit to be considered for anchorage design. The properties of the building are as follows:

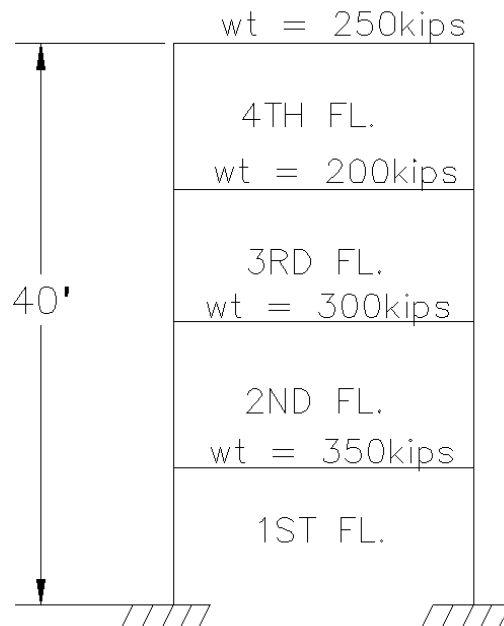
Width of building = 40 ft Roof height = 50 ft

$V = 95$ mph,

Topographic factor $K_{zt} = 1$,

Ground elevation factor $K_e = 1$

- (A) 0.15 kips
(B) 0.25 kips
(C) 0.5 kips
(D) 0.9 kips
7. The design base shear of a 4 story building is calculated as 500 kips. The average story height is 10ft. The fundamental time period of the building is 0.4s. The effective seismic weights of each floor is given below. Calculate the seismic lateral force acting on the 3rd floor of the building.

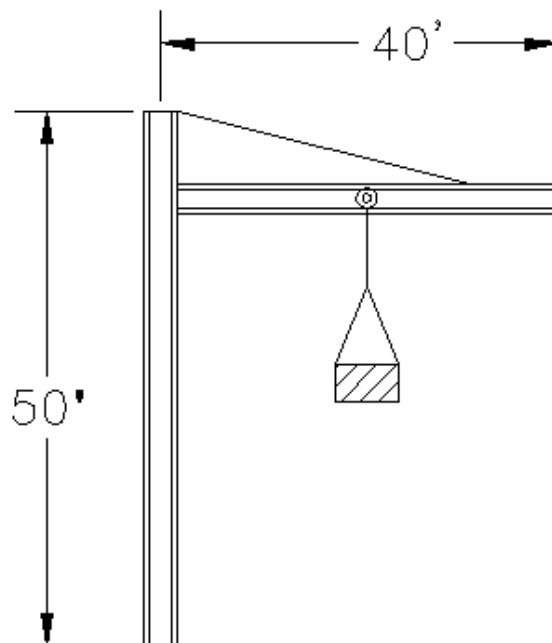


- (A) 118 kips
(B) 125 kips
(C) 200 kips
(D) 250 kips

8. Which of the following is not true about the response modification coefficient “R” in ASCE 7-16 Table 12.2-1

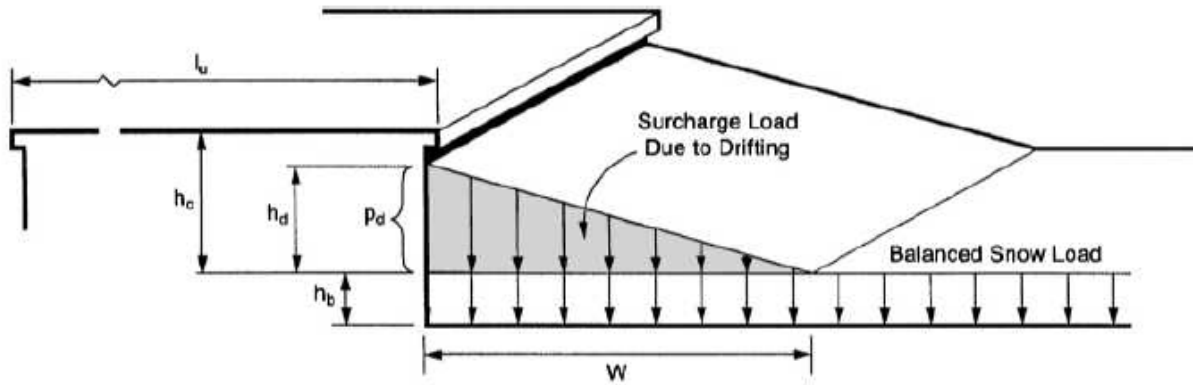
- (A) Forces transferred from the upper system to the lower system shall be increased by multiplying by the ratio of the higher response modification coefficient to the lower response modification coefficient,
- (B) Seismic Loads scaled down using the response modification factor is to be used for LRFD load combinations,
- (C) Higher value of “R” indicates that the system is weaker and has lower resistance to lateral loads,
- (D) The value of the response modification coefficient, R, used for design in the direction under consideration shall not be greater than the least value of R for any of the systems used in that direction.

9. The figure below shows a mini crane that can pick and land a maximum weight of 1ton at a maximum radius of 40ft. The flexural capacity of the girder and column is at 100% utilization at that load and radius. If the hoisting motor, cables and all other components have sufficient residual capacity, what is the estimated maximum weight that the crane can pick and land safely at a minimum operating radius of 5 ft?



- (A) 2 kips
- (B) 8 kips
- (C) 12 kips
- (D) 16 kips

10. Calculate the maximum intensity of leeward drift surcharge ' p_d ' for the lower roof shown below. The ground snow load for the area is 30 psf. Vertical separation between the two roofs is 15ft and the length of the upper roof is 50ft. The snow importance factor of the building is 1.



- (A) 30 psf
 (B) 36.75 psf
 (C) 40.5 psf
 (D) 44.75psf
11. The maximum moment acting on a single lane bridge girder due to truck load is 500 kip*ft. The maximum moment due to design lane load is 600 kip*ft. What is the total unfactored moment including any impact factor per AASHTO LRFD bridge design specifications?
- (A) 1100 kip*ft
 (B) 1265 kip*ft
 (C) 1385 kip*ft
 (D) 1518 kip*ft

12. What is the total lateral stress acting on the retaining wall below with uniform surcharge 50psf?

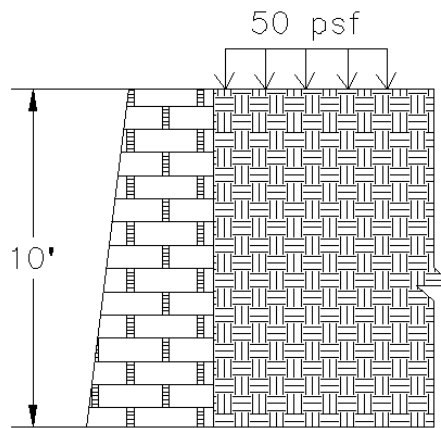
Density, $\gamma = 115$ pcf

Angle of internal friction, $\phi=30$ degree

Cohesion $c=0$

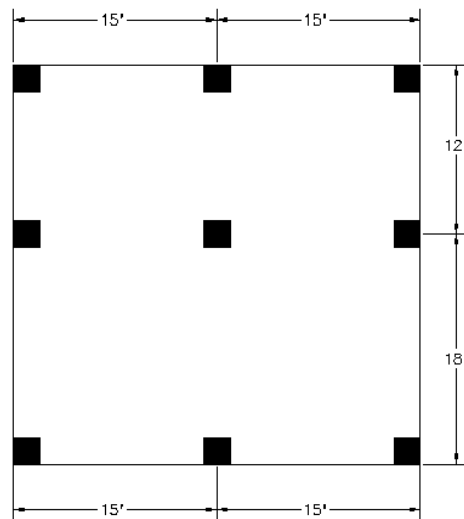
Depth of water table is below the wall

Neglect Wall friction



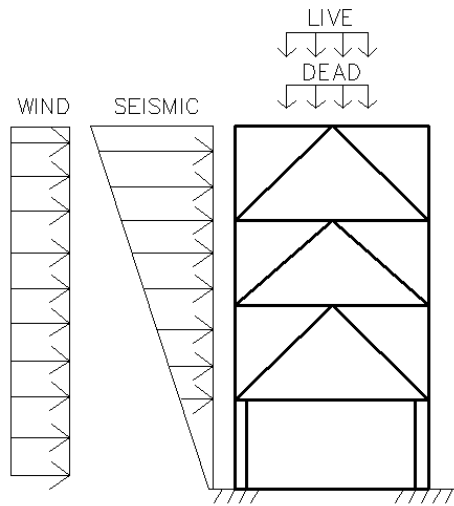
- (A) 1150.5 lb/ft
- (B) 2081.25 lb/ft
- (C) 2120.25 lb/ft
- (D) 3150.4 lb/ft

13. The plan below shows a typical framing plan of a three story building. The dead load on each floor is 60psf and the superimposed dead load is 30 psf. The live load on each floor is 50 psf. What is the total unfactored gravity load acting on the interior column at foundation level?

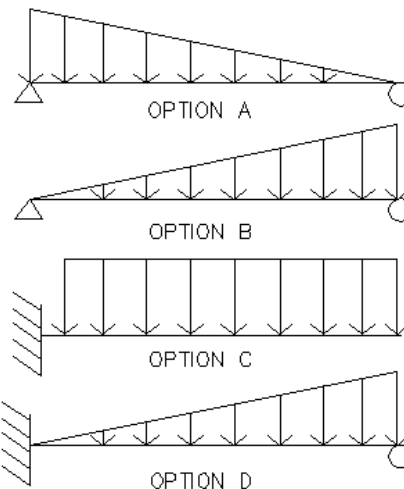


- (A) 55 kips
- (B) 82.5 kips
- (C) 94.5 kips
- (D) 115 kips

14. For the building shown below, which of the following load combinations must be considered to calculate the governing factor of safety for stability against overturning?

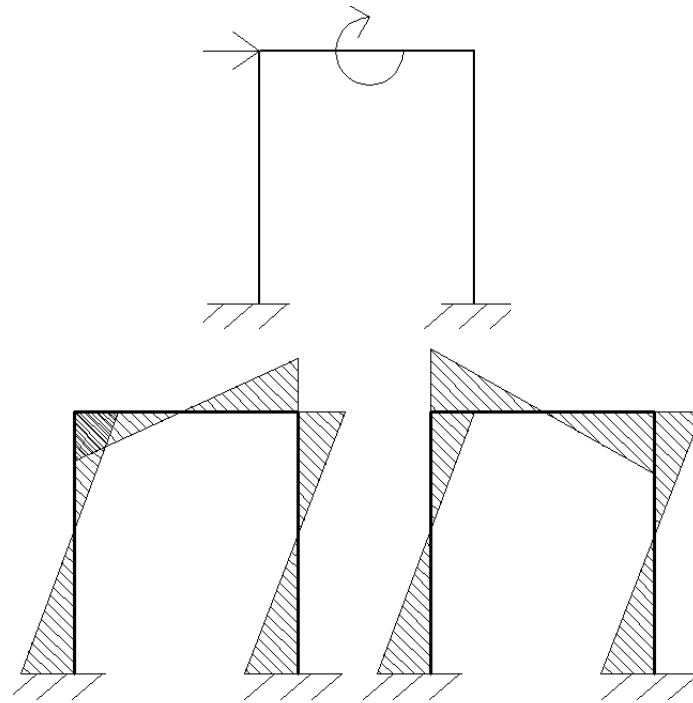


- (A) Dead + Live + Wind
 (B) Dead + Seismic + Wind
 (C) Dead + Seismic or Wind
 (D) Dead + Live + Seismic + Wind
15. For a single span beam, the equation for bending moment diagram is given as $M = 13 + 27x - \frac{2x^3}{3}$ where “x” is the distance from the left support. Which of the following diagrams describes the beam and the loading?



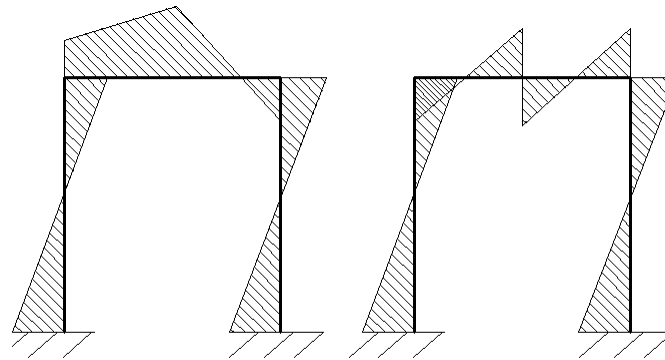
- (A) Option A
 (B) Option B
 (C) Option C
 (D) Option D

16. The frame below has a lateral load acting on the top and an external clockwise moment acting on the center of the beam. Select the correct bending moment diagram.



OPTION A

OPTION B

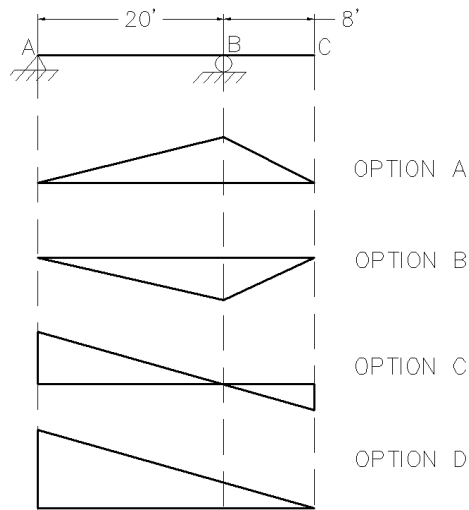


OPTION C

OPTION D

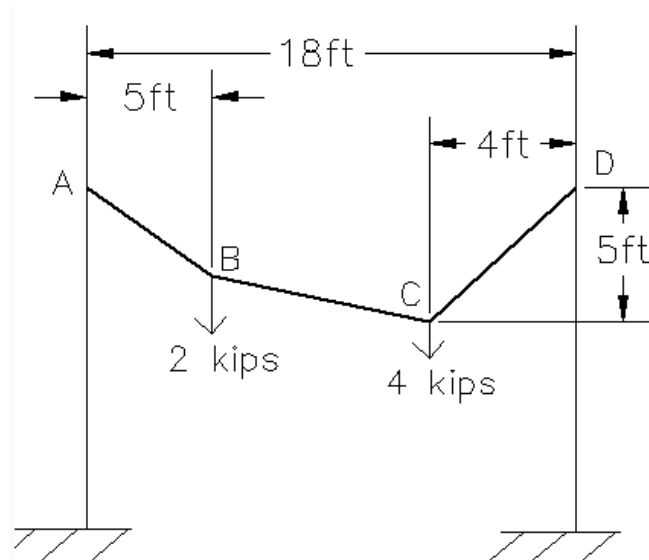
- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

17. For the overhanging beam shown below, select the correct influence line diagram for maximum reaction at A, when a point load passes from point A to point C



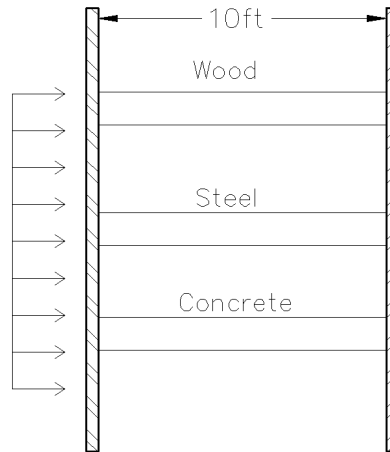
- (A) Option A
(B) Option B
(C) Option C
(D) Option D

18. A cable profile is shown below. The cables are supporting concentrated loads at points B and C. The sag at point C is 5ft. What is the total tension in the cable at section AB? The weight of the cable can be neglected.

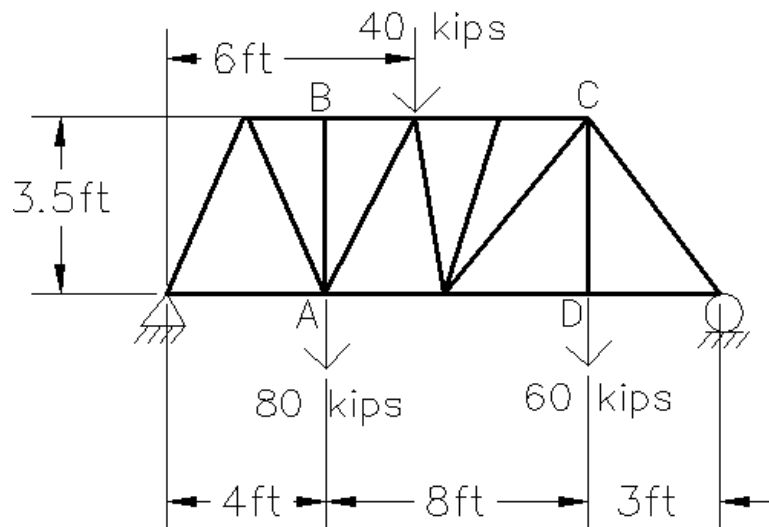


- (A) 2.5 kips
(B) 3.75 kips
(C) 4.5 kips
(D) 5 kips

19. Three horizontal braces are arranged as shown below. The modulus of elasticity of wood is 1500 ksi, modulus of elasticity of concrete is 3500 ksi and modulus of elasticity of steel is 29000 ksi. What is the total horizontal axial stiffness of the system? Cross section area of each brace is 15 in².

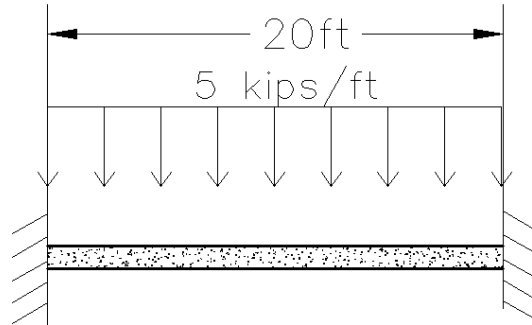


- (A) 3050 kip/in
 (B) 3450 kip/in
 (C) 4000.75 kip/in
 (D) 4250 kip/in
20. For the simply supported truss shown below, calculate the force acting in members AB and CD.



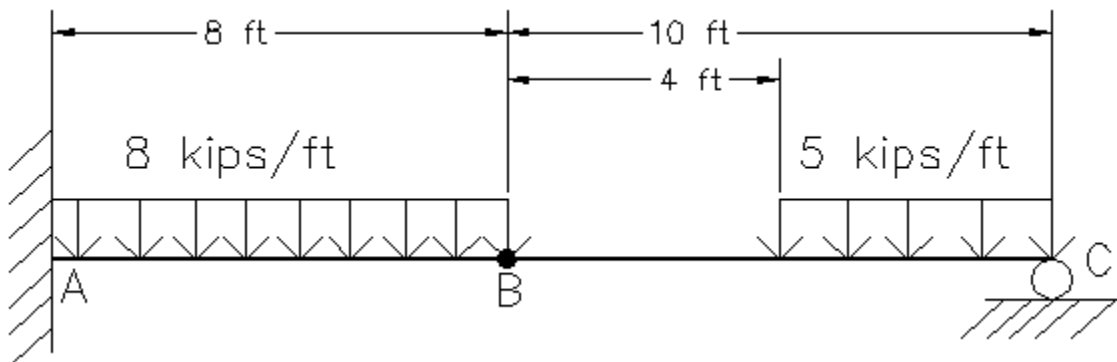
- (A) AB = 0 CD = 0
 (B) AB = 80 kips CD = 60 kips
 (C) AB = 0 kips CD = 60 kips
 (D) AB = 80 kips CD = 0 kips

21. A Reinforced Concrete beam spans between two columns. The span of the beam is 20ft. The only load acting on the beam is a uniform distributed load of 5 kips/ft (factored load) throughout the span. The effective depth 'd' of the beam is 18 inches. What is the critical factored shear force to be considered for design of the beam?



- (A) 40 kips
- (B) 42.5 kips
- (C) 45 kips
- (D) 50 kips

22. For the beam system shown below with a hinge at point B, what is the maximum shear force at hinge location B? Follow general beam sign convention.

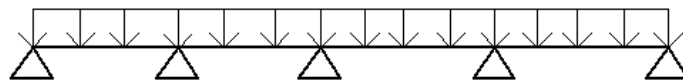


- (A) 0 kips
- (B) -9 kips
- (C) +9 kips
- (D) -29 kips

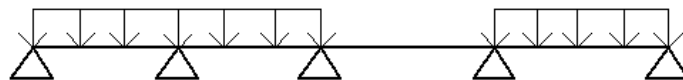
23. In a continuous span beam system , point load is applied directly to the shear center of the beam section. What is true about the load effect and internal forces?

- (A) Beam will only be subjected to pure shear,
- (B) Beam will be subjected to shear and torsion,
- (C) Beam will be subjected to bending moments and shear without torsion,
- (D) Net shear displacement will be zero.

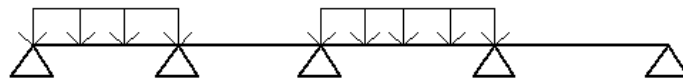
24. Four span continuous beam is subjected to uniformly distributed loadings in different spans as shown below. Which loading configuration produces maximum negative moment in an internal support?



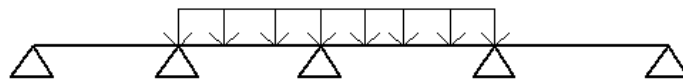
Option A



Option B



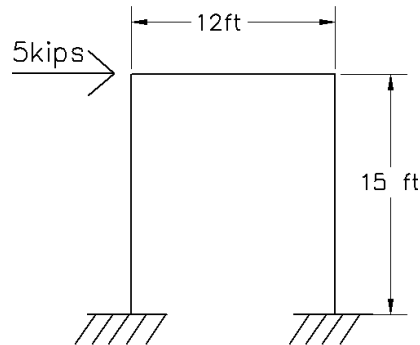
Option C



Option D

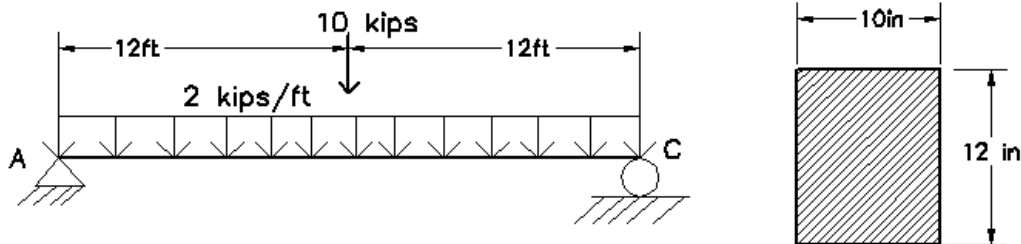
- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

25. What is the maximum bending moment acting on the beam of the frame loaded as shown below? The point of inflection can be assumed at $0.55H$ from the base where H is the height of the column.



- (A) 15 kip*ft
- (B) 17 kip*ft
- (C) 20 kip*ft
- (D) 37.5 kip*ft

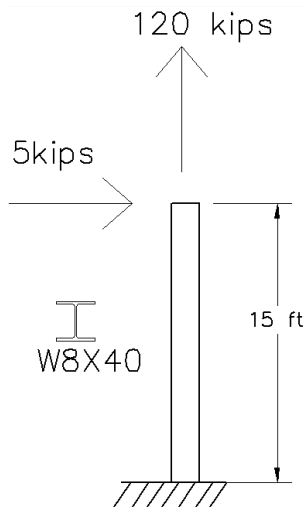
26. For the beam shown below, what is the maximum flexural stress acting on the section? The dimensions of the rectangular cross section is 10in X 12 in.



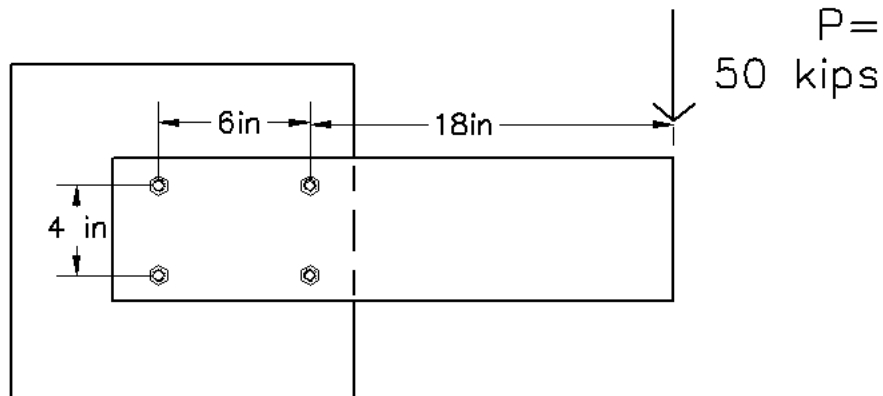
- (A) 9.5 ksi
- (B) 10.2 ksi
- (C) 11.95 ksi
- (D) 13.1 ksi

27. A W8X40 cantilevered column is subjected to a tensile load of 120 kips. A lateral load of 5 kip acts at the top causing bending about the stronger axis. The column is 15 ft tall. What is the maximum compressive stress acting at the fibres of the column section?

- (A) 0 ksi
(B) 5.5 ksi
(C) 8 ksi
(D) 15 ksi



28. The figure below shows a bolted connection that is supporting a 50 kip load. The load is eccentric to the connection geometry creating additional torsion in the connection. What is the maximum shear load on an individual bolt?

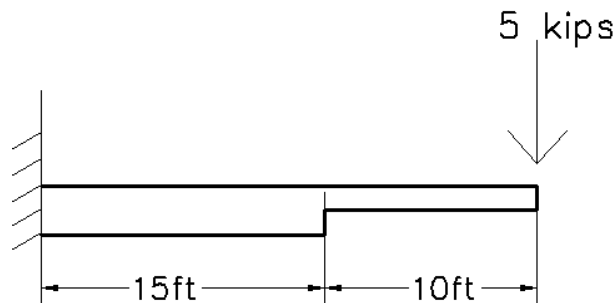


- (A) 12.5 kips
(B) 40.4 kips
(C) 73.1 kips
(D) 83.5 kips

29. For a complete state of stress in a three dimensional body , how many independent stress components are there?

- (A) Two normal components and one shear component,
- (B) Three normal components and three shear components,
- (C) Three normal components and six shear components,
- (D) Two normal components and four shear components.

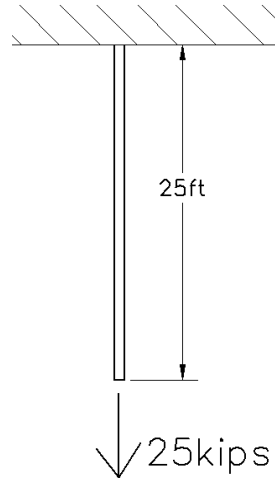
30. A cantilevered beam is loaded at the tip as shown below. The heavier first 15 ft length of the beam has a moment of inertia 1000 in^4 and the lighter 10ft length has a moment of inertia of 500 in^4 . What is the deflection at the tip of the beam? $E = 29000 \text{ ksi}$



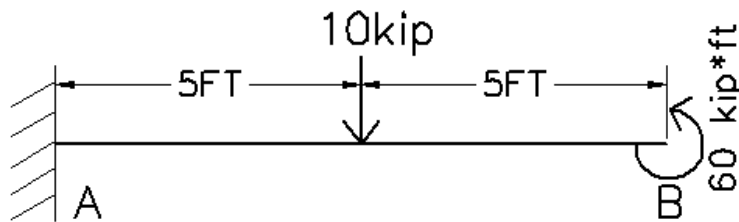
- (A) 1 in
- (B) 1.35 in
- (C) 1.65 in
- (D) 2 in

31. A hanger rod of 1.85 inch diameter is measured to be 25ft long when the ambient temperature was 100°F. During winter, a load of 25 kips is applied to the rod and the ambient temperature is 0°F. What is the deflection at the tip of the rod?
Modulus of Elasticity = 29000 ksi Coefficient of thermal expansion = 0.000007 /°F.

- (A) 0.11 inch downwards
(B) 0.21 inch downwards
(C) 0.11 inch upwards
(D) 0.21 inch downwards

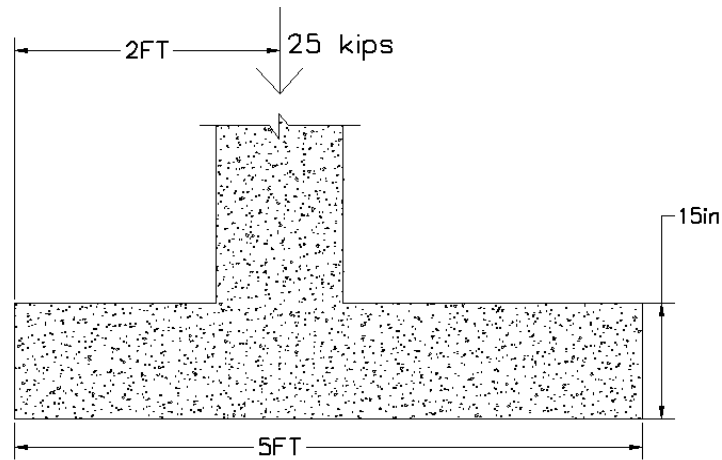


32. What is the deflection at the free end of the cantilever beam due to the loads shown below?
 $I = 50 \text{ in}^4$ $E = 29000 \text{ ksi}$

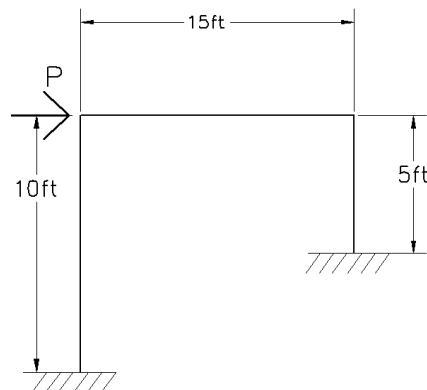


- (A) 1.2 in
(B) 2.33 in
(C) 3.1 in
(D) 4.8 in

33. The figure below shows a 5 ft X 5ft square footing. The column load including self weight of the column acting on the footing is 25 kips. What is the maximum bearing pressure? The unit weight of concrete = 0.15 kcf.



- (A) 1 ksf
(B) 1.6 ksf
(C) 1.8 ksf
(D) 2.8 ksf
34. For the frame shown below, the plastic moment capacity of the section is 150 kip*ft. What is the value of lateral collapse load P?



- (A) 60 kip
(B) 70 kip
(C) 80 kip
(D) 90 kip

35. Which of the following constructions does not mandate special inspections?

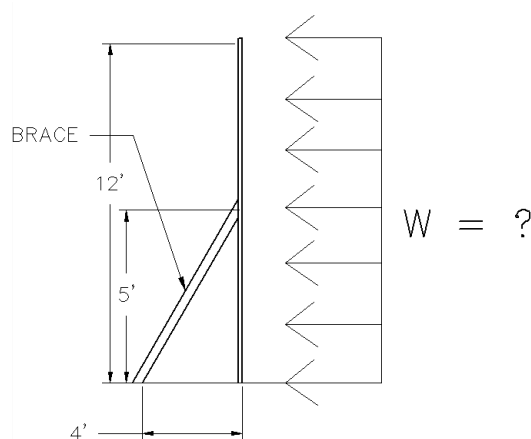
- (A) An isolated concrete footing for a 4 story building that is supported on rocky soil,
- (B) An over reinforced continuous footing that is constructed with 3 ksi concrete and is analyzed to have sufficient structural strength with 25% lesser rebar,
- (C) An under reinforced footing that is constructed with 3 ksi concrete and is analyzed to have sufficient structural strength with 25% lesser compressive strength of concrete,
- (D) A 5 inch thick slab on grade that serves as a foundation to the diesel generator

36. Which of the following does not typically require special inspection per IBC chapter 17?

- (A) Installation of prestressing tendons,,
- (B) Soil bearing capacity for shallow foundations ,
- (C) Thickness of fireproofing applications,
- (D) Operations of a concrete batching plant.

37. A 12' tall wall is braced at a height of 5 feet as shown in the figure below. The wall is subjected to a uniformly distributed lateral load W lbs/ft. The bottom of the wall can be considered pin jointed. The allowable axial strength of the brace is 20 kips. What is the maximum allowable lateral load ' W ' on the brace?

- (A) 550 lbs/ft
- (B) 750 lbs/ft
- (C) 850 lbs/ft
- (D) 950 lbs/ft



38. The table shows allowable loads for shoring props.

A 550 prop is to be used to shore a floor that is subjected to a 100 psf dead load, 75 psf construction load and 15 psf load due to weight of all formwork and falsework. If the floor height is 15ft, what is the maximum spacing of the props that can be used to shore the floor?

- (A) 4 ft X 4 ft,
- (B) 4.5 ft X 4.5 ft
- (C) 5 ft X 5 ft
- (D) 5.5 ft X 5.5 ft

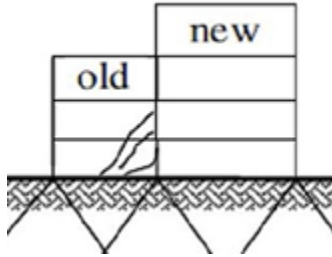
Permitted prop loads [kN]

Prop height [m]	Eurex 20				
	250	300	350	400	550
5.5					
5.4					
5.3					
5.2					
5.1					
5.0					
4.9					
4.8					
4.7					
4.6					
4.5					
4.4					
4.3					
4.2					
4.1					
4.0					
3.9					
3.8					
3.7					
3.6					
3.5					
3.4					
3.3					
3.2					
3.1					
3.0					

39. A multi level guard rail is installed on a single point adjustable suspension scaffold. What is the design moment acting on the midrail per OSHA regulations, if it is welded between two posts spaced 5 ft apart.

- (A) 50 lbs*ft
- (B) 75 lbs*ft
- (C) 100 lbs*ft
- (D) 150 lbs*ft

40. A new building is to be constructed next to an old building causing stress overlaps under their foundations. Which of the following is not a remedial measure to counter settlements in the old building due to the new construction?



- (A) Improve the strength parameters of the soil underneath,
- (B) Using pile foundations for the new building,
- (C) Placing the foundation of the new building at a depth above the foundation of the old building,
- (D) Designing lateral load bearing systems of the new building to be placed at the center of the new building. .

41. The particle size distribution of an inorganic soil sample is as follows:

Liquid Limit = 55
Plastic Limit = 17

Determine the AASHTO classification of the soil sample.

- (A) A-2-7 (14)
- (B) A-2-7 (22)
- (C) A-7-6(14)
- (D) A-7-6(22)

Soil Size (mm)	Fraction retained on sieve
<0.005	0.42
0.005-0.05	0.11
0.05-0.075	0.12
0.075-2.0	0.35
>2.0	0

42. An excavation trench is to be filled using soil from a borrow pit nearby. The in situ density of the soil in the borrow pit is 1600 kg / m^3 and the in situ water content is 10 %. For the fill area, the required dry density of the soil must be 1800 kg/m^3 and the water content must be 12 %. What is the volume of soil to be excavated from the borrow pit area if the volume of the trench to be filled is $20,000 \text{ m}^3$?

- (A) 22500 m^3
- (B) 24750 m^3
- (C) 26250 m^3
- (D) 28000 m^3

43. A pile cap is about to be cast in a coastal area and it is exposed occasionally to tides. The design strength of concrete is 4000 psi. Which of the following mixes should be chosen to cast the pier?

- (A) Type I cement, sand, aggregates, water, chemical accelerator,
- (B) Type I cement, sand, aggregates, water, super plasticizer,
- (C) Type II cement, sand, aggregates, water, chemical accelerator,
- (D) Type II cement, sand, aggregates, water, super plasticizer

44. 250 ft^3 of concrete mix is to be produced. The mix proportion by weight is 1: 2 :3. Six gallons of water is to be used per bag of cement. How many bags of cement will be required for the job if the target air content in the mix is 5%?

Density of cement = 195 lbf/ft^3

Density of fine aggregate = 160 lbf/ft^3

Density of coarse aggregate = 164 lbf/ft^3

Weight of cement bag = 94 lbs

- (A) 40 bags
- (B) 50 bags
- (C) 60 bags
- (D) 70 bags

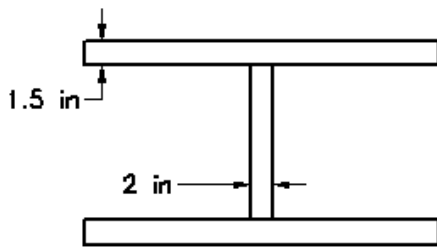
45. What is the development length of #8 compression reinforcement with yield strength 60 ksi for 6000 psi normal weight concrete?

- (A) 16 in
- (B) 18 in
- (C) 20 in
- (D) 30 in

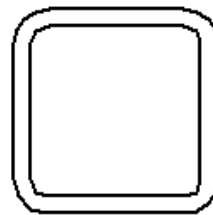
46. Which of the following is true about effects of temperature on different properties of steel?

- (A) Increase in temperature increases the yield and ultimate strength of steel.
- (B) Increase of temperature reduces the ductility and hardness of steel.
- (C) Temperature has no effect on the modulus of commonly used structural steel.
- (D) At lower temperatures, the stiffness of steel members increases.

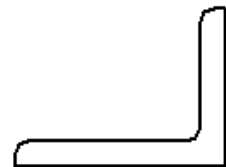
47. Match the preferred material specification with the appropriate steel section per AISC specifications.



(I) Plate Girder



(II) HSS 8X8



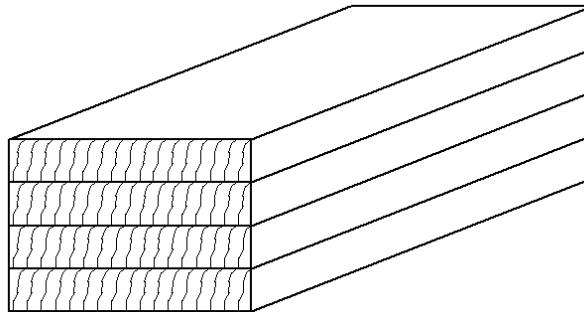
(III) L8X6

- | | | |
|--------------|----------|------------|
| (A) I - A992 | II- A572 | III - A709 |
| (B) I - A572 | II- A500 | III - A36 |
| (C) I - A992 | II- A572 | III - A618 |
| (D) I - A500 | II- A572 | III - A36 |

48. A Wide flange section has 1 inch cover plates welded to the bottom flange. The moment of inertia of the composite section is 2238 in^4 . The location of the centroid from the top flange is 9.8 in. What is the maximum flexural stress acting on the cover plate due to a moment of 350 kip*ft?



- (A) 11.6 ksi
(B) 18.4 ksi
(C) 22.3 ksi
(D) 27.5 ksi
49. Determine the ASD adjusted modulus of elasticity for stability calculations of a 5 DF L1 with 4 laminations. The member is primarily subjected to axial loads. Service temperature = 120°F and service condition is wet.



- (A) 794 ksi
(B) 832 ksi
(C) 1060 ksi
(D) 1128 ksi

50. Which of the following is not true about the property of mortar used in masonry construction?

- (A) During construction, the temperature of the mortar mix should always be maintained less than 120°F and above 40°F.
- (B) The strength of the mortar should always be greater than the strength of the individual masonry units,
- (C) Type M mortars have high strength and is used for heavy duty applications,
- (D) Type N and Type O mortars have high workability.

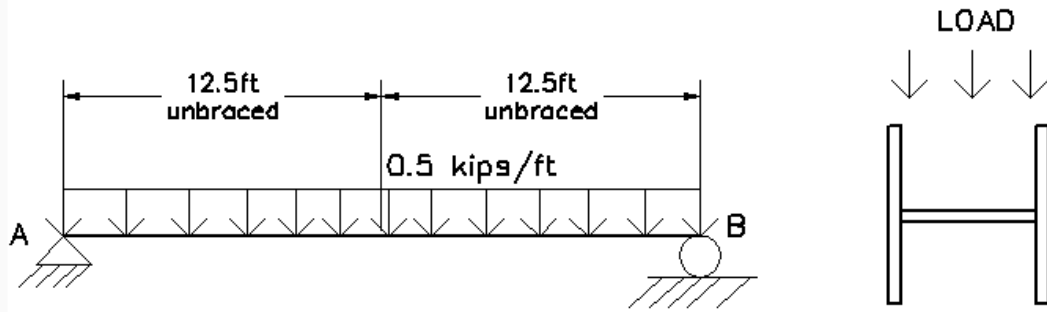
51. Which of the following is not true about concrete compressive strength tests methods and sampling?

- (A) A strength test shall be the average of strengths of at least two 6X12 or 4X8 cylinders.
- (B) Samples of the concrete shall be made at the point of delivery of concrete.
- (C) Concrete cylindrical specimens should be tested per ASTM C39
- (D) Sampling of concrete cylinders should be done per ASTM C172

52. A concrete masonry assembly is constructed using individual units of strength 2000 psi tested per ASTM C90 and using Type N mortar. Thickness of mortar beds is ½ inch. What is the specified compressive strength 'f'm' of the masonry assembly?

- (A) 1500 psi
- (B) 1750 psi
- (C) 2000 psi
- (D) 2650 psi

53. A simply supported steel beam has a span of 25 ft and is braced at the midpoint. The beam is subjected to a uniformly distributed service live load of 0.5 kip/ft and bends about the minor axis. What is the lightest Gr. 50 section that will satisfy the flexural strength requirements?



- (A) W18X46
 - (B) W18X50
 - (C) W18X55
 - (D) W18X60
54. Determine the major axis nominal shear strength of a Gr 50 Channel section C15X40 beam. No transverse stiffeners are used. $E = 29000$ ksi



- (A) 168 kips
- (B) 192 kips
- (C) 234 kips
- (D) 263 kips

55. A fully grouted reinforced masonry beam is subjected to a shear force of 13 kips. If a single legged #4 bar is used as shear reinforcement, what is the shear stress carried by the shear reinforcement?

Section properties:

$$f'_m = 1500 \text{ psi.}$$

$$M/Vdv > 1$$

$$b = 7.625 \text{ in}$$

$$dv = 24 \text{ in}$$

$$d = 20 \text{ in}$$

- (A) 19.2 psi
(B) 21 psi
(C) 22.5 psi
(D) 27.5 psi
56. A 15 ft long simply supported 24F-V1 SP/SP timber beam has thickness of 4 inches and depth of 16 inches. If the beam is subjected to a uniformly distributed load of 100 lb/ft, what is the volume factor 'Cv' of the beam?
- (A) 0.8
(B) 1
(C) 1.01
(D) 1.21
57. A four span continuous one way slab is constructed using Gr 60 rebar and lightweight concrete with $f'_c = 4000$ psi and unit weight 110 pcf. The interior spans are 21 ft in length and exterior spans are 17 ft in length. What is the minimum required thickness of the slab?

- (A) 6 in
(B) 8.5 in
(C) 9 in
(D) 10 in

58. The factored shear to be resisted by a RCC beam section is 70 kips. The dimensions of the beam is 12inX24 in and the compressive strength of concrete is $f'_c = 4000$ psi. What is the maximum permissible spacing of stirrups per the provisions of ACI 318-14 for the beam section? Effective cover for tension bars = 2.5in. Concrete is normal weight.
- (A) 6 in
 - (B) 9 in
 - (C) 12 in
 - (D) 24 in
59. A cantilevered W8X40 column is subjected to compression load. The effective length in stronger - X direction, $KL_x = 28$ ft. The effective length in weaker - Y direction, $KL_y = 14$ ft. The material of the column is A992 Gr 50. What is the axial compressive strength of the column to be used in strength design?
- (A) 100 kips
 - (B) 150 kips
 - (C) 250 kips
 - (D) 321 kips
60. A vertical steel post consists of an angle section loaded in compression, away from its centroid. The designer wants to neglect the effect of eccentricity of the load and evaluate the member as axially loaded only. Which of the following conditions should be met so that the designer's assumption is valid.
- I) Slenderness should be less than 200,
 - II) Members should be loaded on the same leg on both ends,
 - III) The length of legs of the angle should be equal.
- (A) III
 - (B) I and II
 - (C) I and III
 - (D) I, II and III

61. The effective length of a W14X82 Grade 50 column is 15ft in both directions. What is the maximum allowable axial compression it can carry, if it is also subjected to a strong axis flexure of 80 kip*ft and weak axis flexure of 40 kip*ft. Lateral torsional buckling is prevented by bracing the compression flanges only.

$E = 29000 \text{ ksi}$

- (A) 125 kips
- (B) 201 kips
- (C) 233 kips
- (D) 489 kips

62. Select Structural DFL 2X6 studs are braced effectively about the weaker axis with gypsum boards on both sides as shown below with adequate fastener arrangement. What is the maximum allowable axial compression per stud?

Section properties

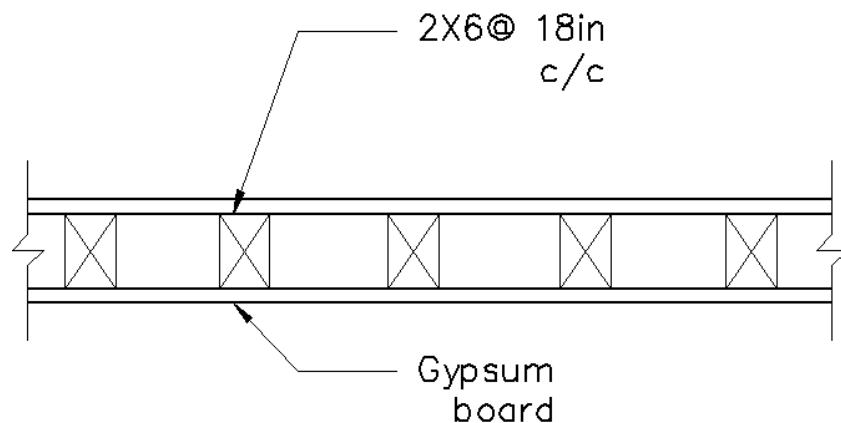
$$C_d = 1$$

$$C_f = 1.15$$

$$C_m = C_t = C_i = 1$$

$$C_p = 0.33 \text{ about strong axis}$$

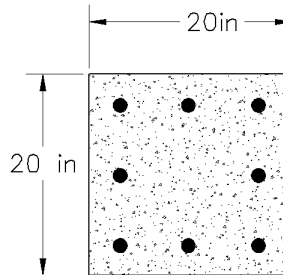
$$C_p = 1 \text{ about weak axis}$$



- (A) 2 kips
- (B) 5 kips
- (C) 8 kips
- (D) 16 kips

- 63.** What is the design shear strength of the column section shown below? The column is subjected to a factored axial load of 150 kips.

$$f'_c = 4000 \text{ psi} \quad f_y = 60 \text{ ksi}$$



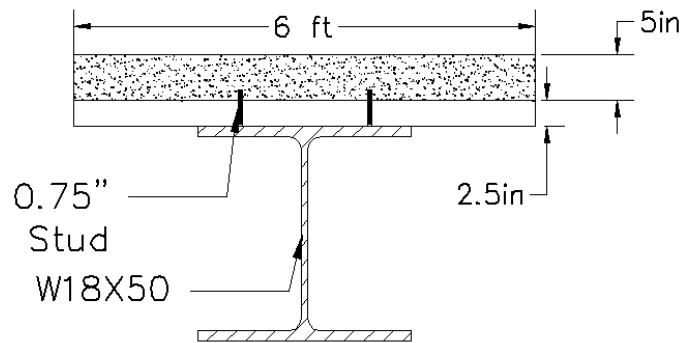
- (A) 25 kips
(B) 38.3 kips
(C) 45.1 kips
(D) 52.5 kips
- 64.** A building is to be designed in a high seismic region. Which of the following systems is most likely to provide the highest level of redundancy and performance for resisting lateral seismic loads?
- (A) Specially reinforced cantilevered columns,
(B) Flat concrete slabs with drop panels,
(C) Reinforced masonry shear walls ,
(D) Combined moment frame and shear wall.

65. A simply supported composite beam with span 28 ft has ribs of deck perpendicular to the beam span as shown below. The effective width of the slab is 6 ft and two 0.75 inch diameter headed studs are used per rib. For full composite action, what is the total number of studs required? The beam is subjected to uniformly distributed load.

Nominal shear strength of stud, $Q_n = 15$ kips.

W18X50 Gr. 50

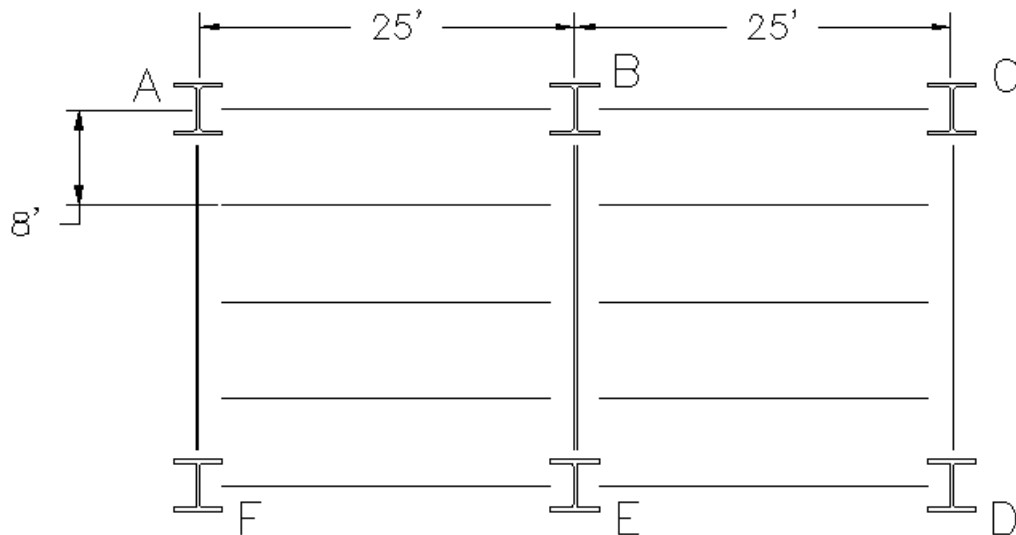
Slab $f'_c = 4$ ksi



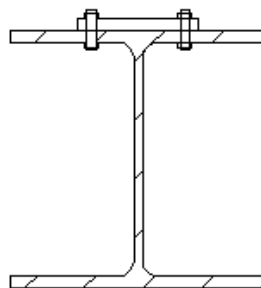
- (A) 36
(B) 49
(C) 98
(D) 116
66. A 1 inch diameter headed steel stud is embedded in a composite concrete slab with strength $f'_c = 6000$ psi and modulus of elasticity, $E = 4400$ ksi. There are two studs per deck rib located at strong positions. Assuming that the failure of steel does not control, what is the nominal strength of the headed anchor?

- (A) 51.2 kips
(B) 55.9 kips
(C) 63.8 kips
(D) 72.3 kips

67. Partial framing plan of a building is shown below. The spandrel beam AB supports a 13 ft tall, 8 in thick wall with unit weight 80 lbf/ft^2 . The floor dead load and live loads are 75 lbf/ft^2 and 100 lbf/ft^2 . Assuming simple connections and full lateral support of compression flange, what is the most economical section that satisfies strength requirements for beam AB? Material used is Gr. 50.

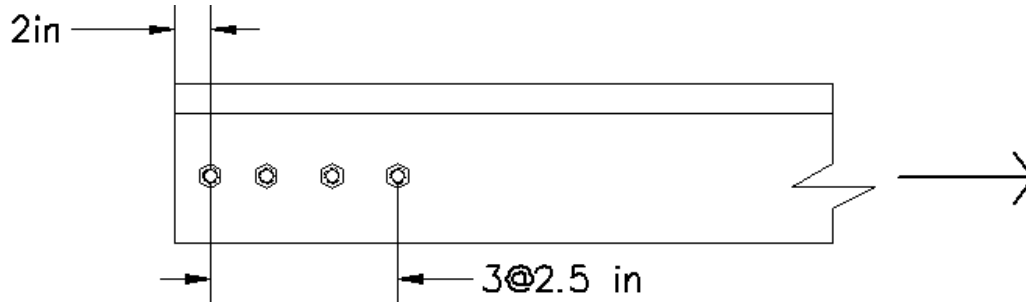


- (A) W12X30
 (B) W12X40
 (C) W12X45
 (D) W12X53
68. A $3/4$ inch thick plate is to be bolted on top of a W section flange as shown below using standard size holes. The diameter of the bolt is $1/2$ inches. What is the minimum clear distance from the edge of the plate to the edge of the hole?

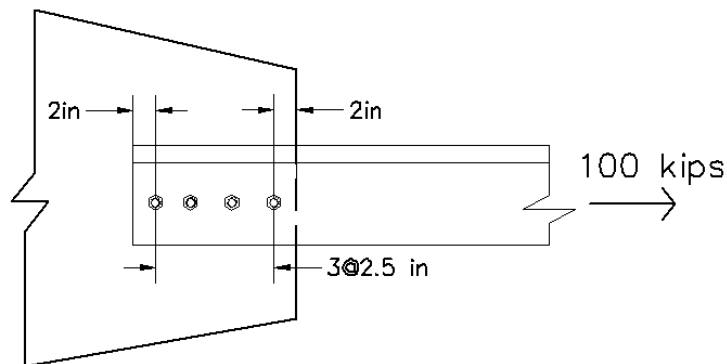


- (A) $15/32$ in
 (B) $9/16$ in
 (C) $5/8$ in
 (D) $3/4$ in

69. An angle section 5X3X1/2 used as a tension member, is connected to a plate with four 0.75 inch diameter bolts as shown below. What is the governing allowable bolt bearing and tearout strength of an individual bolt if the holes are standard sizes? Yield strength of angle is 36 ksi and tensile strength is 58 ksi. Deformation at bolt hole at service load is a design consideration.



- (A) 20 kips
(B) 22 kips
(C) 26 kips
(D) 28 kips
70. Determine the required minimum thickness of a gusset plate that connects an angle with four bolts as shown below. Assume tensile yielding of the plate governs. Yield strength of plate is 36 ksi and tensile strength is 58 ksi. The loads shown are service loads.

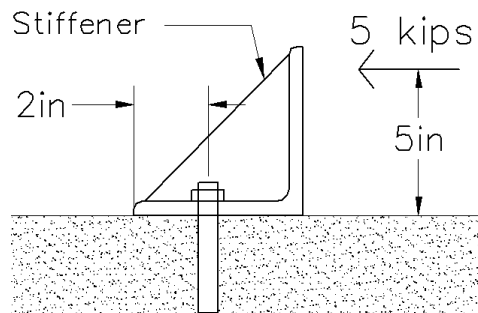


- (A) 1/2 in
(B) 5/8 in
(C) 3/4 in
(D) 1 in

71. For a beam to column moment connection, which of the following factors most significantly affects the rotational stiffness of the connection?

- (A) The grade of steel used for beams and columns,
- (B) The length of the fillet welds along the beam web,
- (C) The diameter of bolts used in the flange connection plates,
- (D) The thickness and size of flange plates or angles,

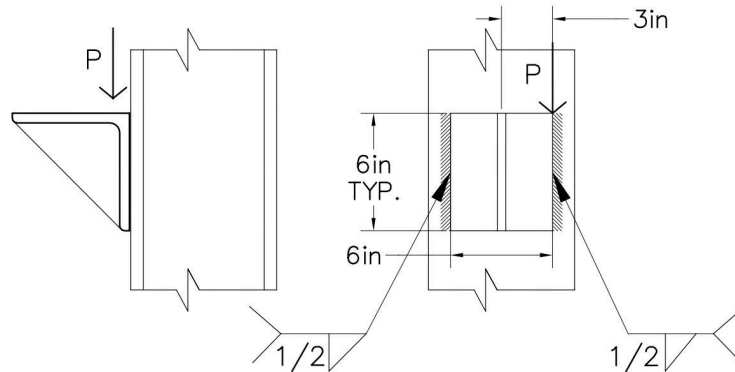
72. An angle section is connected to a concrete slab with a bolt as shown below. The length of the angle section is 4 inches. Stiffener plates are welded to the angle section at the ends to make the connection rigid. What is the minimum tension acting in the bolts?



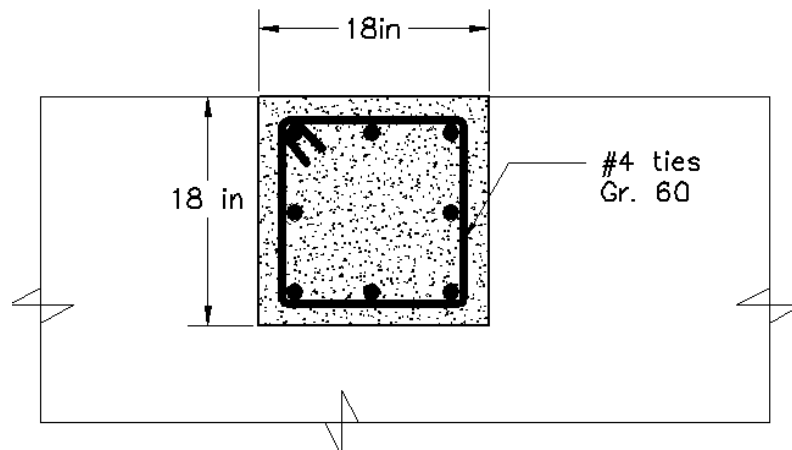
- (A) 5 kips
- (B) 8.5 kips
- (C) 10.2 kips
- (D) 12.5 kips

73. A bracket is welded on the face of the column as shown below. Using instantaneous center of rotation method, what is the maximum allowable load that 'P' that the bracket can carry, assuming that the welds govern the failure of connection.

$F_{exx} = 70 \text{ ksi}$



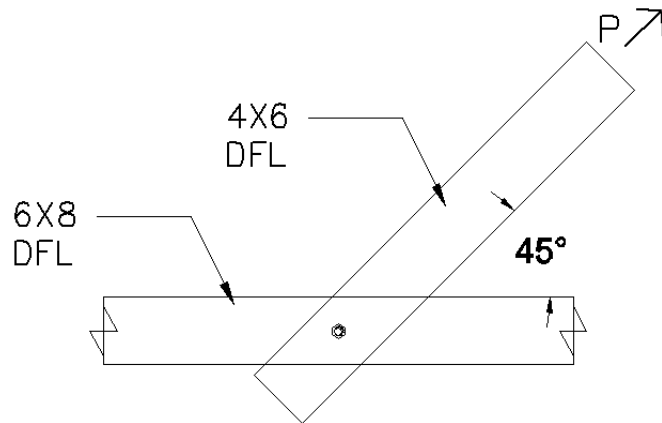
- (A) 41.2 kips
(B) 48.2 kips
(C) 62.4 kips
(D) 71.3 kips
74. Plan of an exterior column with slab on three sides is shown below. The grade of concrete of both column and slab is 4 ksi. The thickness of the slab is 8 in and the structure is not in a critical seismic zone. What is the minimum number of Gr. 60 #4 ties required inside the slab- column joint ?



- (A) 1
(B) 2
(C) 3
(D) 4

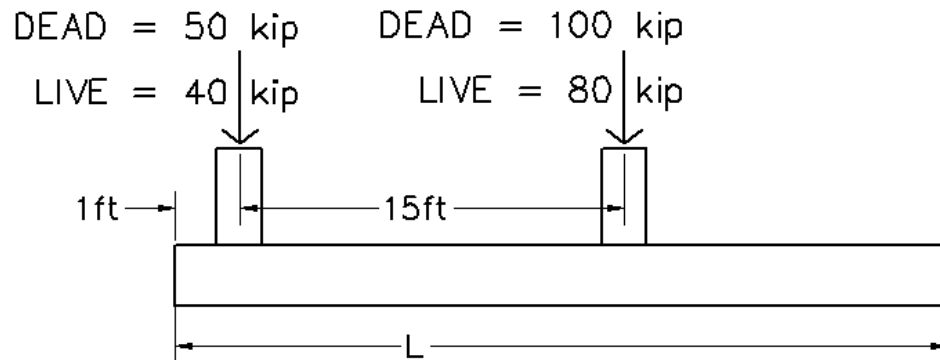
75. A 6X8 DFL member is connected to a 4X6 DFL member with a single 1 inch diameter bolt. The connection is loaded at an angle 45° as shown below. What is the maximum allowable load 'P' in the connection due to the loading angle? Material of the bolt and bearing doesn't govern failure. All other adjustment factors are equal to 1.

- (A) 1390 lbs
- (B) 1522 lbs
- (C) 1826 lbs
- (D) 2260 lbs

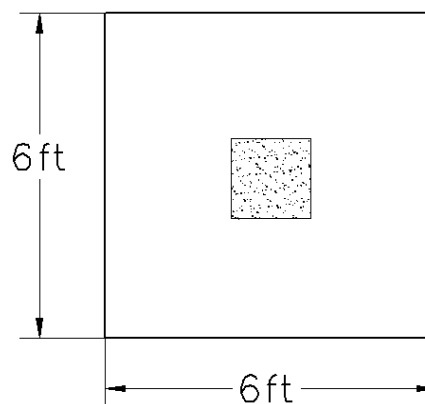


76. Which of the following statements is not true about controlling differential settlements and its effects in a building with shallow foundations?
- (A) Planting trees and adequate vegetation around the perimeter of the property help stabilize and regulate soil moisture and control differential settlement,
 - (B) Proper surface and subsurface drainage systems in the property prevent water accumulation and minimize water related differential settlement.
 - (C) Expansion joints play a crucial role in controlling the effects of differential settlement,
 - (D) Soil compaction and stabilization reduces the risk of differential settlement.

77. A combined footing supports two columns with axial loads as shown below. What should the length of footing 'L' be such that there is uniform bearing pressure under the combined footing.



- (A) 18 ft
(B) 22 ft
(C) 25 ft
(D) 32 ft
78. The plan of a square RCC footing is shown below. The dimensions of the column at the center of the footing is 18 in X 18 in. The effective depth of the footing is 21 in and there are no shear reinforcements. What is the design two way shear strength of the footing?
 $f'_c = 4 \text{ ksi}$ $\lambda = 1$

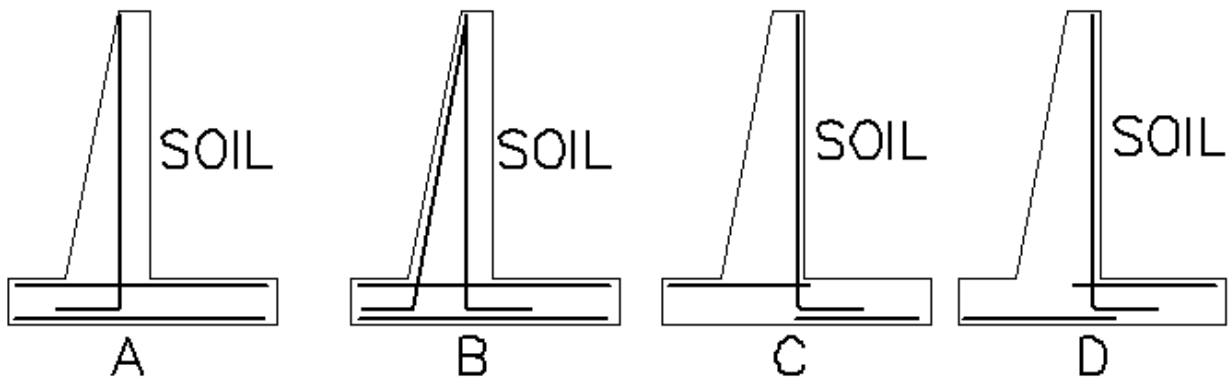


- (A) 418 kips
(B) 505 kips
(C) 622 kips
(D) 828 kips

79. A concrete caisson is driven 20 ft inside the ground. The thrust acting on the caisson is 75 kips and the uplift acting is 45 kips. The thrust and uplift act in the caisson separately. The allowable end bearing is 12 ksf and the allowable skin friction is 300 psf. What is the minimum required diameter of the caisson ?

- (A) 2 ft
- (B) 2.5 ft
- (C) 2.75 ft
- (D) 3 ft

80. The cross section of a cantilevered retaining wall that is retaining soil on the vertical face of the wall is shown below. What is the correct reinforcement layout for the wall ?



- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

Solutions

[Click here for Problems](#)

1. Problem

Correct Answer: (C) **264.6 lb/ft**

Explanation:

Weight of steel deck = 5 psf

Density of lightweight concrete = 115 lb/ft³ {Ref ACI 318-14 page 33}

Calculate the weight of slab per unit area

Slab weight = 3 inches * 115 lb/ft³ = 28.75 psf

Calculate Floor load acting on steel beam spaced at 6ft c/c

{Ref AISC steel manual 15th Ed. page 1-29 Table 1-1 for self weight of W8X18}

Floor Load = deck weight + slab weight = (5psf + 28.75 psf) * 6ft = 202.5 lb/ft

Total load acting on beam = self-weight of W8X18 beam + Floor Load

= 18 lb/ft + 202.5 lb/ft

= 220.5 lb/ft

Total Factored Dead Load (used in live load combination) = 220.5 lb/ft * 1.2 = **264.6 lb/ft**

2. Problem

Correct Answer: (C) The expected service life of the structure

Explanation:

Per IBC 2018 section 1607.11,

Live load reduction is not permitted for heavy loads exceeding 100 psf,

Live load reduction factor depends on the tributary area in equation 16-23,

Live load reduction is not permitted in structures designed for storage.

There are no clauses discussing the service life of structures and corresponding live load reduction.

Hence option C is the correct answer as it does not affect the reduction.

3. Problem

Correct Answer: (B) 5760 lbs

Explanation:

Per IBC 2018 table 1607.1, page 368, minimum live load to be considered for a school classroom is 40psf.

Calculate the total expected load from given information and check against this minimum requirement.

Total expected live load due to students and furniture

$$= (10+2)*200 \text{ lb} + 2000 \text{ lb}$$

$$= 4400 \text{ lb}$$

Calculate minimum design load per IBC.

$$\text{Area} = 12\text{ft} * 12\text{ft} = 144 \text{ ft}^2$$

$$\text{Load} = 40\text{psf} * 144\text{ft}^2$$

$$= 5760 \text{ lbs}$$

Minimum requirement governs over expected load, hence correct answer is **5760 lbs**

4. Problem

Correct Answer: (C) The load factors for permanent live loads and construction loads are different in ASD methodology.

Explanation:

The statement is false because the load factor is 1 in ASD methodology for both permanent live loads and construction loads.

5. Problem

Correct Answer: (C) Dead + Wind

Explanation:

Dead load is the only guaranteed permanent load that resists uplift when wind load is acting

Hence to evaluate uplift, the combination 'Dead + Wind' will be most critical and is the correct answer.

6. Problem

Correct Answer: (D) 0.9 kips

Explanation:

$$V = 95 \text{ mph}$$

Velocity pressure at any height is given by ASCE 7-16 equation 26.10-1

$$q_h = 0.00256 K_z K_{zt} K_d K_e V^2$$

Exposure category B can be assumed when no information is given.

From chapter 26, for 50ft, Exposure B, Velocity pressure coefficient $K_z = 0.81$

From Table 26.6-1, Wind directionality factor, $K_d = 0.85$ for rooftop equipment

$$q_h = 0.00256 * 0.81 * 1 * 0.85 * 1 * 95^2 = 15.9 \text{ psf}$$

Lateral pressure on equipment is given by ASCE 7-16 equation 29.4-2 page 322

$$F_h = q_h (GC_r) A_f$$

$$A_f = 6\text{ft} * 5\text{ft} = 30\text{ft}^2 \text{ (Projected area of vertical face of unit)}$$

$$\text{Width of building} = 40\text{ft} \quad h = 50\text{ft}$$

$$0.1 * B * h = 200\text{ft}^2$$

Since projected area A_f is less than $(0.1B * h)$, $GC_r = 1.9$

$$F_h = 15.9\text{psf} * 1.9 * 30\text{ft}^2$$

$$F_h = 906 \text{ lbs} \approx \mathbf{0.9 \text{ kips}}$$

7. Problem

Correct Answer: (A) 118 kips

Explanation:

$$\text{Base shear } V = 500 \text{ kips}$$

1st floor seismic weight $w_1 = 350$ kips, height from ground level $h_1 = 10\text{ft}$

2nd floor seismic weight $w_2 = 300$ kips, height from ground level $h_2 = 20\text{ft}$

3rd floor seismic weight $w_3 = 200$ kips, height from ground level $h_3 = 30\text{ft}$

4th floor seismic weight $w_4 = 250$ kips, height from ground level $h_4 = 40\text{ft}$

For time period less than 0.5 seconds, $k = 1$ from section 12.8.3

Per ASCE 7-16, page 102 section 12.8.3, the lateral force distribution is given by equation 12.8-11 and equation 12.8-12

$$F_x = C_{vx} * V$$

$$C_{vx} = \frac{w_x h_x^k}{\sum w_i h_i^k}$$

$$\sum w_i h_i^k = 350 * 10 + 300 * 20 + 200 * 30 + 250 * 40 = 25500 \text{ kip} * \text{ft}$$

For 3rd floor,

$$C_{v3} = \frac{w_3 h_3^k}{\sum w_i h_i^k} = \frac{200*30}{25500} = 0.235$$

$$F_3 = C_{v3} * V = 0.235 * 500 \text{ kips} = 117.5 \text{ kips}$$

Thus, force acting 3rd floor = 117.5 kips ≈ **118 kips**

8. Problem

Correct Answer: (C) Higher value of “R” indicates that the system is weaker and has lower resistance to lateral loads,

Explanation:

Per ASCE 7-16 section 12.2.3.1, option A is true

Option B is true

Option D is true per section 12.2.3.3

Option C is incorrect because higher value of R denotes robust system and higher resistance to lateral loads.

9. Problem

Correct Answer: (D) **16 kips**

Explanation:

At maximum radius of 40ft, the girder and column is controlled by flexure and limited by 1 ton weight. Per the question, all other components have sufficient residual capacity. So we need to calculate moment acting on the girder when 1 ton load is at maximum radius.

$$\text{Moment acting due to 1ton load at 40 ft} = 1 \text{ ton} * 40 \text{ ft} = 2 \text{ kips} * 40 \text{ ft} = 80 \text{ kip} * \text{ft}$$

We need to equate this moment to the moment caused by a load at 5ft

$$P * 5\text{ft} = 80 \text{ kip*ft}$$

$$P = 80 \text{ kip*ft} / 5\text{ft}$$

$$= 16 \text{ kip}$$

Hence the crane can safely pick and land a weight of **16 kip** at the minimum radius 5ft.

10. Problem

Correct Answer: (D) 44.75 psf

Explanation:

From ASCE 7-16, section 7.7.1, page 59, calculate the density of the snow,

Ground snow load $p_g = 30 \text{ psf}$

From equation 7.7-1, density of snow,

$$\gamma = 0.13 * p_g + 14 = 0.13 * 30 + 14 = 17.9 \text{ pcf}$$

Length of upper roof $l_u = 50\text{ft}$

Importance factor $I_s = 1$

Get drift height from page 61, figure 7.6-1

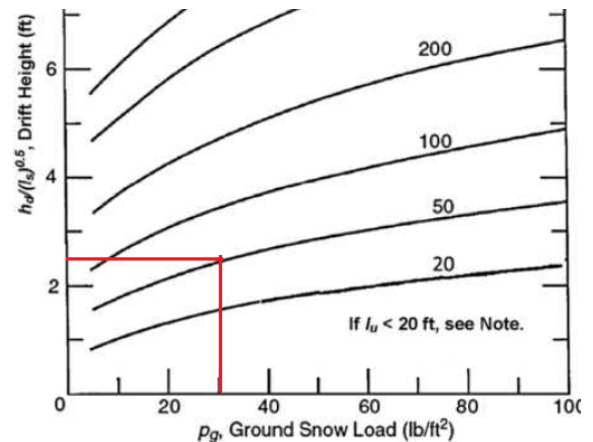
For $l_u = 50\text{ft}$ and $p_g = 30 \text{ psf}$

Since $I_s = 1$, we get $h_d = 2.5 \text{ ft}$

Thus, maximum intensity of drift surcharge,

from section 7.7.1

$$p_d = \gamma * h_d = 17.9 \text{ pcf} * 2.5 \text{ ft} = 44.75 \text{ psf}$$



11. Problem

Correct Answer: (D) 1518 kip*ft

Explanation:

Per AASHTO LRFD 2017, page 3-31 section 3.6.2, dynamic load allowance i.e. impact factor will not applied to design lane load and will only be applied to truck load with a value 33%

Thus, moment due to truck load with impact factor

$$= 1.33 * 500 \text{ kip*ft}$$

$$= 665 \text{ kip*ft}$$

Thus total moment with impact/dynamic allowance

= truck load with impact + design lane load without impact

$$= 665 \text{ kip*ft} + 500 \text{ kip*ft} = 1265 \text{ kip*ft}$$

Also, for single lane load, from table 3.6.1.1.2-1 page 3-21, multiple presence factor = 1.2

$$\text{Thus total unfactored moment} = m * 1265 \text{ kip*ft} = 1.2 * 1265 \text{ kip*ft} = \mathbf{1518 \text{ kip*ft}}$$

12. Problem

Correct Answer: (B) 2081.25 lb/ft

Explanation:

Given, wall face is vertical and wall friction can be neglected

Use Rankine's equation for active earth pressure coefficient

$$K_a = \tan^2(45 - \phi/2)$$

$$= \tan^2(45 - 30/2)$$

$$= 0.333$$

Dry density of soil, $\gamma = 115 \text{ pcf}$

Surcharge, $q = 50 \text{ psf}$

Height of wall, $H = 10 \text{ ft}$

Total load due to surcharge and earth pressure

$$= K_a * q * H + \frac{1}{2} * K_a * \gamma * H^2 = 0.333 * 50 \text{ psf} * 10 \text{ ft} + \frac{1}{2} * 0.333 * 115 \text{ pcf} * 10 \text{ ft} * 10 \text{ ft} = \mathbf{2081.25 \text{ lb/ft}}$$

13. Problem

Correct Answer: (C) 94.5 kips

Explanation:

Tributary area for interior column

$$= (15 \text{ ft}/2 + 15 \text{ ft}/2) * (12 \text{ ft}/2 + 18 \text{ ft}/2)$$

$$= 225 \text{ ft}^2$$

From each floor, load shared by interior column

$$\text{Dead load} = 60 \text{ psf} * 225 \text{ ft}^2 = 13500 \text{ lbs} = 13.5 \text{ kips}$$

$$\text{Superimposed dead load} = 30 \text{ psf} * 225 \text{ ft}^2 = 6750 \text{ lbs} = 6.75 \text{ kips}$$

Live load = $50\text{psf} \times 225\text{ft}^2 = 11250\text{ lbs} = 11.25\text{kips}$

Total load from 3 floors = $3 \times (13.5 + 6.75 + 11.25)\text{ kips} = \mathbf{94.5\text{ kips}}$

14. Problem

Correct Answer: (C) Dead + Seismic or Wind

Explanation:

Code does not require us to combine seismic and wind loads at the same time. Live load helps to counter overturning so should not be used in calculation against uplift and overturning. Hence the correct combination is option C, dead + seismic or wind (whichever governs between seismic and wind)

15. Problem

Correct Answer: (D)

Explanation:

The given equation of bending moment is a cubic equation. So, the shear force equation should be parabolic and the loading diagram should be linear. Hence Option C is ruled out. Also, when we put $x = 0$ in the equation, $M = 13$. Hence the left support should be able to resist moments and is fixed. This rules out options A and B.

Option D is the answer.

16. Problem

Correct Answer: (D)

Explanation:

An external moment at the center of the beam brings an abrupt change to the moment diagram. Option A and B have no changes in the slope of the moment diagram and are ruled out. Option C shows bending moment decreasing gradually at the point of application of

moment. This is true if there was a point load, not a moment. Option D shows an abrupt decrease in the moment at the center of the beam and is the correct answer.

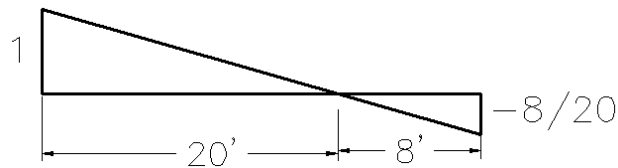
17. Problem

Correct Answer: (C)

Explanation:

The influence lines for reactions and internal forces for section between supports should be constructed as for beam without overhangs and extended along the overhangs,

For reaction at A, the maximum reaction will occur when the load is directly on top of point A and will be zero when the load is directly on top of point B. The value of ordinate at point A will be equal to 1. Thus the influence line diagram is



The correct answer is Option C.

18. Problem

Correct Answer: (B) 3.75 kips

Explanation: For cables,

The moment at any point can be taken as zero.

Let vertical and horizontal reactions at point A be A_x and A_y

Let vertical and horizontal reactions at point D be D_x and D_y

Taking moment at point A = 0

$$D_y \cdot 18\text{ft} = 4 \text{ kips} \cdot (18\text{ft} - 4\text{ft}) + 2 \text{ kips} \cdot 5\text{ft}$$

$$D_y = 66/18 = 3.67 \text{ kips}$$

Sum of vertical forces = 0

$$D_y + A_y = 2 \text{ kips} + 4 \text{ kips}$$

$$A_y = 6 \text{ kips} - 3.67 \text{ kips} = 2.33 \text{ kips}$$

Taking moments about point C = 0

$$D_x * 5\text{ft} = D_y * 4\text{ft}$$

$$D_x = D_y * (4 \text{ ft} / 5\text{ft}) = 2.936 \text{ kips}$$

Taking sum of horizontal forces = 0

$$A_x = D_x = 2.936 \text{ kips (in opposite direction)}$$

Now, total tension in section AB

$$T = \sqrt{A_x^2 + A_y^2} = \sqrt{2.936^2 + 2.33^2} = \mathbf{3.75 \text{ kips}}$$
 Correct answer is option B.

19. Problem

Correct Answer: (D) 4250 kip/in

Explanation:

Here, when horizontal load acts, the deformation in each braces will be equal. Thus the total stiffness of the system can be assumed as three springs in parallel and will be the sum of each individual stiffness

$$\text{Total stiffness, } K = K_1 + K_2 + K_3$$

For each brace,

$$\text{Length, } L = 10\text{ft}$$

$$\text{Area, } A = 15 \text{ in}^2$$

Modulus of elasticity 'E' is variable depending on material property

For Wood,

$$K_1 = (E_{\text{wood}} * A) / L = 1500 \text{ ksi} * 15 \text{ in}^2 / \{10\text{ft} * 12(\text{in}/\text{ft})\} = 187.5 \text{ kip/in}$$

For Steel,

$$K_2 = (E_{\text{steel}} * A) / L = 29000 \text{ ksi} * 15 \text{ in}^2 / \{10\text{ft} * 12(\text{in}/\text{ft})\} = 3625 \text{ kip/in}$$

For Concrete,

$$K_3 = (E_{\text{concrete}} * A) / L = 3500 \text{ ksi} * 15 \text{ in}^2 / \{10\text{ft} * 12(\text{in}/\text{ft})\} = 437.5 \text{ kip/in}$$

$$\text{Total stiffness } K = 187.5 + 3625 + 437.5 = \mathbf{4250 \text{ kip/in}}$$

Correct answer is option D.

20. Problem

Correct Answer: (C) AB = 0 and CD = 60 kips

Explanation:

Analyze joint B,

There are no external loads and BA is the only vertical member.

Taking sum of vertical forces = 0 , we get

Force in BA = 0

Analyze joint D with an external load of 60 kips

Taking sum of vertical forces = 0

Force in CD = 60 kips (CD is the only vertical member)

Hence the correct answer is option C.

21. Problem

Correct Answer: (B) 42.5 kips

Explanation:

Per ACI 318-14 page 132, section 9.4.3.2, the critical section for shear in beams is at a distance 'd' from the face of support and beams shall be permitted to be designed for shear at that critical section

Effective depth, $d = 18 \text{ inches} = 1.5 \text{ ft}$

Load, $w = 5 \text{ kips/ft}$

Span, $L = 20 \text{ ft}$

Reaction at support = $wL/2$

$= 5 * 20 / 2 = 50 \text{ kips}$

Calculate shear at critical section i.e. at distance 'd'

Shear at 1.5 ft

$= \text{Reaction} - W * d$

$= 50 - 5 * 1.5 = 42.5 \text{ kips}$

Hence design shear is **42.5 kips** and the correct answer is option B. .

22. Problem

Correct Answer: (C) +9 kips

Explanation:

Total load due to 5 kips/ft distributed load = $5 \text{ kips/ft} * 6 \text{ ft} = 30 \text{ kips}$

Location from the hinge of centroid of the force = $4 \text{ ft} + 6 \text{ ft} / 2 = 7 \text{ ft}$

Let R_c be the reaction at point C

At the internal hinge, sum of moments = zero. Considering forces on the right of the hinge,

$$\sum M_B = 0 \quad R_c * 10 \text{ ft} = 30 \text{ kips} * 7 \text{ ft} \quad R_c = 21 \text{ kips}$$

Shear at point B = $-R_c + 30 \text{ kips} = -21 \text{ kips} + 30 \text{ kips} = +9 \text{ kips}$,

Correct answer is option C.

23. Problem

Correct Answer: (C)

Explanation:

When load passes through the shear center of a section, there will be no twisting. Hence there will be no torsion. There will be bending and shear in the span and also some shear deflection. Hence, the correct option is C.

24. Problem

Correct Answer: (B)

Explanation:

Ref to continuous beam formulas in NCEES PE handbook page 253-255

Maximum negative moment occurs in the 2nd support when the first, second and fourth spans are loaded and the third span is unloaded. The value of the moment is $-0.121wl^2$.

Thus correct answer is option B

25. Problem

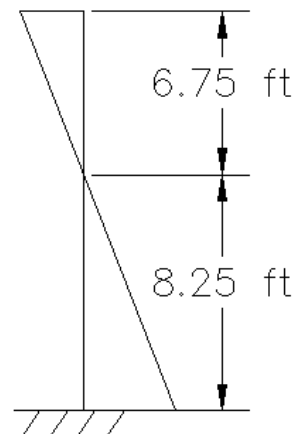
Correct Answer: (B) 17 kip *ft

Explanation:

At the point of inflection, the bending moment is zero.

Due to symmetry, the shear acting on the each fixed support is

$$V = 5 \text{ kips}/2$$



$$V = 2.5 \text{ kips}$$

The point of inflection is at $0.55H$ from the base

$$0.55 \times 15 \text{ ft} = 8.25 \text{ ft}$$

The shear at point of contraflexure = 2.5 kips

For the given loading, the beam moment is maximum at the column joint.

At the top of the column,

$$\text{Moment} = 2.5 \text{ kips} \times (15 \text{ ft} - 8.25 \text{ ft})$$

$$= 16.875 \text{ kip} \cdot \text{ft} \approx \mathbf{17 \text{ kip} \cdot \text{ft}}$$

Correct answer is option B.

26. Problem

Correct Answer: (B) 10.2 ksi

Explanation:

Maximum bending moment acting in the beam

Due to point load at center

$$= P \cdot L / 4$$

$$= 10 \text{ kips} \times 24 \text{ ft} / 4$$

$$= 60 \text{ kip} \cdot \text{ft}$$

Due to UDL

$$= w \cdot L^2 / 8$$

$$= 2 \text{ kips/ft} \times (24 \text{ ft})^2 / 8 = 144 \text{ kip} \cdot \text{ft}$$

$$\text{Total moment} = 60 + 144 = 204 \text{ kip} \cdot \text{ft}$$

Section modulus of rectangular section

$$S = b \cdot h^2 / 6$$

$$S = 10 \text{ in} \times 12 \text{ in} \times 12 \text{ in} / 6$$

$$S = 240 \text{ in}^3$$

Flexural stress

$$= M / S = 204 \text{ kip} \cdot \text{ft} \times 12 (\text{in/ft}) / 240 \text{ in}^3 = \mathbf{10.2 \text{ ksi}}, \text{ Correct answer is option B.}$$

27. Problem

Correct Answer: (D) 15 ksi

Explanation:

Ref AISC steel manual page 1-28 and page 1-29 for section properties of W8X40.

$$\text{Area (A)} = 11.7 \text{ in}^2$$

$$\text{Section modulus (Sx)} = 35.5 \text{ in}^3$$

$$\text{Tensile load (P)} = 120 \text{ kips}$$

$$\text{Moment (Mx)} = 5 \text{ kips} \times 15 \times 12 \text{ in} = 900 \text{ kip}\cdot\text{in}$$

$$\text{Maximum compressive stress due to bending} = Mx/Sx$$

$$= 900 \text{ kip}\cdot\text{in} / 35.5 \text{ in}^3$$

$$= 25.35 \text{ ksi}$$

$$\text{Maximum tensile stress due to tensile load} = P/A$$

$$= 120 \text{ kips} / 11.7 \text{ in}^2$$

$$= 10.26 \text{ ksi}$$

$$\text{Net compressive stress} = 25.35 \text{ ksi} - 10.26 \text{ ksi} = 15.1 \text{ ksi} \approx \mathbf{15 \text{ ksi}}$$

Correct answer is option D.

28. Problem

Correct Answer: (D) 83.5 kips

Explanation:

Ref to NCEES PE handbook page 257, section 4.2.1 for shear in eccentrically loaded bolts

No. of bolts, $n = 4$

Due to symmetry location of centroid of bolt group from bottom left bolt ,

$$x = 3 \text{ in} \quad y = 2 \text{ in}$$

$$\text{Eccentric moment acting on the group, } M = 50 \text{ kips} \times (18 \text{ in} + 6 \text{ in} / 2) = 1050 \text{ kip}\cdot\text{in}$$

Moment of inertia of bolt group (Area of all the bolts is assumed equal)

$$I_x = \sum x^2 = 4 \times 3 \text{ in} \times 3 \text{ in} = 36 \text{ in}^2 \quad I_y = \sum y^2 = 4 \times 2 \text{ in} \times 2 \text{ in} = 16 \text{ in}^2$$

$$I_p = I_x + I_y = 36 + 16 = 52 \text{ in}^2$$

$$\text{Vertical load on each bolt due to direct shear} = 50 \text{ kips} / 4 = 12.5 \text{ kips}$$

$$\text{Vertical load on each bolt due to moment} = (M/I_p) \times x = 1050 / 52 \times 3 = 60.6 \text{ kips}$$

$$\text{Total vertical shear} = 12.5 + 60.6 = 73.1 \text{ kips}$$

$$\text{Horizontal shear on each bolt due to moment} = (M/I_p) \times y = 1050 / 52 \times 2 = 40.4 \text{ kips}$$

$$\text{Total shear} = \sqrt{(73.1 \text{ kips})^2 + (40.4 \text{ kips})^2} = \mathbf{83.5 \text{ kips}}$$

Correct answer is option D.

29. Problem

Correct Answer: (B)

Explanation:

For a 2D body subjected to plane stress, there are three independent stress components - two normal and one shear.

For a 3D body, the total number of independent stress components are six - three normal and three shear.

Thus the correct answer is option B.

30. Problem

Correct Answer: (C) 1.65 in

Explanation:

The deflection at any point is given by the moment of area of the M/EI diagram at that point.

The M/EI diagram is obtained by drawing a bending moment diagram for the beam and dividing the ordinates by EI .

For heavier section, $EI = 1000E$

For lighter section, $EI = 500E$

At support. Max moment = 5 kips*25ft = 1500 kip*in

$$1500/1000E = 1.5/E$$

At 10 ft from the tip, Moment = 5 kips* 120 in = 600 kip*in

For heavier section, $600/1000E = 0.6/E$

For lighter section, $600/500E = 1.2/E$

$$A1 = 0.5*(1.5-0.6)/E*180 = 81/E$$

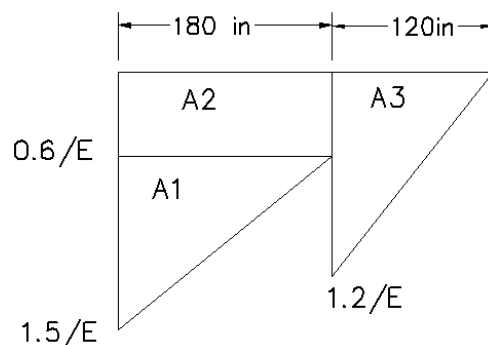
$$X1 = 180*2/3 + 120 = 240 \text{ in}$$

$$A2 = 0.6/E*180 = 108/E$$

$$X2 = 180*1/2 + 120 = 210 \text{ in}$$

$$A3 = 0.5*1.2/E*120 = 72/E$$

$$X3 = 2/3*120\text{in} = 80\text{in}$$



Moment of all areas about the tip

$$= A1*x1 + A2*x2 + A3*X3$$

$$= 81/E*240 + 108/E*210 + 72/E*80$$

$$= 47880/E$$

$$= 47880/29000$$

$$= 1.65 \text{ in}$$

Correct answer is option C.

31. Problem

Correct Answer: (C) 0.11 inch upwards

Explanation:

Ref NCEES PE handbook page 28-29 for stresses, strains and thermal deformations,

$$\text{Area, } A = \pi/4 * (1.85 \text{ in})^2 = 2.69 \text{ in}^2$$

$$\text{Length, } L = 25 \text{ ft}$$

$$\alpha = 0.000007/^{\circ}\text{F}$$

$$E = 29000 \text{ ksi}$$

$$\Delta T = 0 - 100^{\circ}\text{F} = -100^{\circ}\text{F}$$

Change in length due to thermal strain

$$= \alpha * \Delta T * L = 0.000007 * (-100) * 25 \text{ ft} * 12 (\text{in/ft}) = -0.21 \text{ in (decrease in length)}$$

Change in length due to load = PL/AE

$$= 25 \text{ kip} * 25 \text{ ft} * 12 (\text{in/ft}) / (2.69 \text{ in}^2 * 29000 \text{ ksi}) = 0.096 \text{ in (increase in length)}$$

$$\text{Total deflection at tip} = 0.096 \text{ in} - 0.21 \text{ in} = -0.11 \text{ in (upwards)}$$

Correct answer is option C.

32. Problem

Correct Answer: (B) 2.33 in

Explanation:

Calculate deflections using the moment area method. Deflection at free end is given by the first moment of area of M/EI diagram between fixed end and free end, about the free end.

Moment diagram due to point load

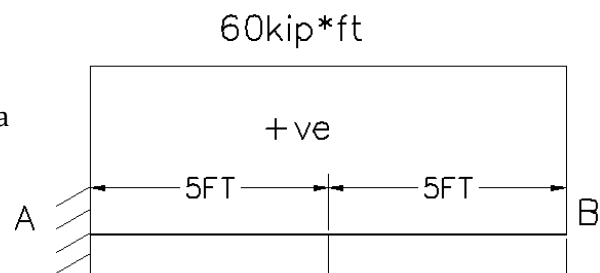
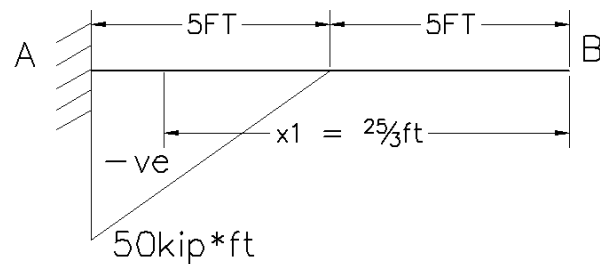
Area of M/EI diagram

$$A1 = -0.5 * 5 * 50/EI$$

$$A1 = -125/EI$$

Centroid from B

$$x1 = 25/3 \text{ ft}$$



Moment diagram due to moment at free end

Area of M/EI diagram

$$A_2 = 10 \cdot 60 / EI$$

$$A_2 = 600 / EI$$

Centroid from B

$$x_2 = 5 \text{ ft}$$

Deflection = Sum of moment of areas due to A1 and A2 (all units in inches and kips)

$$= A_1 \cdot x_1 + A_2 \cdot x_2$$

$$= (-125 / EI \cdot 25 / 3 + 600 / EI \cdot 5) \cdot (12 \text{ in/ft})^3$$

$$= 1958.33 / EI \cdot (12 \text{ in/ft})^3$$

$$= 1958.33 / (29000 \cdot 50) \cdot (12 \text{ in/ft})^3$$

$$= \mathbf{2.33 \text{ in}}$$

Correct answer is option B.

33. Problem

Correct Answer: (C) 1.8 ksf

Explanation:

$$L = 5 \text{ ft}$$

$$B = 5 \text{ ft}$$

Calculate the eccentricity,

$$e = 5 \text{ ft} / 2 - 2 \text{ ft} = 0.5 \text{ ft}$$

Eccentricity is less than $L/6$, so all areas of footing are in compression.

Calculate pressure due to eccentric axial load

$$q_1 = P/A + P \cdot e / (B \cdot L^2 / 6)$$

$$= 25 \text{ kips} / (5 \text{ ft} \cdot 5 \text{ ft}) + (25 \text{ kips} \cdot 0.5 \text{ ft}) / (5 \text{ ft} \cdot 5 \text{ ft} \cdot 5 \text{ ft} / 6)$$

$$= 1.6 \text{ kips/ft}^2$$

Calculate pressure due to self weight of footing

$$q_2 = 0.15 \text{ kips/ft}^3 \cdot 15 \text{ in} \cdot (1/12 \text{ ft/in})$$

$$q_2 = 0.188 \text{ kips/ft}^2$$

$$\text{Total bearing pressure} = q_1 + q_2$$

$$= 1.6 \text{ kips/ft}^2 + 0.188 \text{ kips/ft}^2$$

$$= 1.788 \text{ kips/ft}^2 \approx \mathbf{1.8 \text{ ksf}}$$

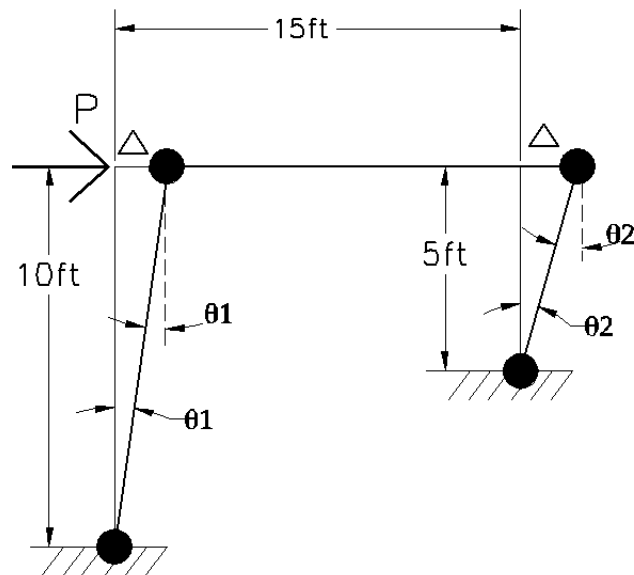
Correct answer is option C.

34. Problem

Correct Answer: (D) 90 kip

Explanation:

The collapse of the structure is due to the formation of four plastic hinges as shown below



Let θ_1 and θ_2 be the rotations at the taller and shorter columns

$$\Delta = \theta_1 \cdot 10\text{ft}$$

$$\Delta = \theta_2 \cdot 5\text{ft}$$

$$\theta_2 = \theta_1 \cdot 10\text{ft} / 5\text{ft} = 2\theta_1$$

$$\text{Total rotation of plastic hinge} = \theta_1 + \theta_1 + \theta_2 + \theta_2 = 6\theta_1$$

External work done = internal work done

$$P \cdot \Delta = M_p \cdot 6\theta_1$$

$$P = M_p \cdot 6\theta_1 / (10\text{ft} \cdot \theta_1)$$

$$P = 6M_p / 10\text{ft} = 6 \cdot 150\text{ kip}\cdot\text{ft} / 10\text{ft} = \mathbf{90\text{ kip}}$$
 Correct answer is option D

35. Problem

Correct Answer: (C)

Explanation:

Ref to IBC 2018 page 418, section 1705.3.

Isolated concrete spread footings of buildings 3 story or less are exempt and Option A is incorrect.

Between Options B and C, Option C is the one that requires concrete strength, f'_c of $0.75 \times 3 \text{ ksi} = 2.25 \text{ ksi}$. Since the footing design is based on concrete less than 2500 psi, it is exempt from special inspections and is the correct answer.

36. Problem

Correct Answer: (D)

Explanation:

All options require special inspections per IBC Chapter 17 except batching plant operations unless specifically required by the contract documents. Hence it is the correct answer.

37. Problem

Correct Answer: (C) 850 lbs/ft

Explanation:

Total lateral load due to uniform load = $12 \text{ ft} \times W$

Point of action of uniform lateral load at mid height = $12 \text{ ft} / 2 = 6 \text{ ft}$

Angle of inclination of the brace = $\tan^{-1}(5 \text{ ft} / 4 \text{ ft}) = 51.34 \text{ degree}$

Horizontal component of the allowable axial force on the brace

= $20 \text{ kips} \times \cos(51.34 \text{ deg}) = 12.5 \text{ kips}$

This component must be equal to the horizontal reaction at brace point.

Taking sum of moments at the base of wall due to lateral load and horizontal component of brace reaction = 0

$$12 \text{ ft} * W * 6 \text{ ft} - 12.5 \text{ kips} * 5 \text{ ft} = 0$$

$$W = 12.5 \text{ kips} * 5 \text{ ft} / (12 \text{ ft} * 6 \text{ ft})$$

$$W = 0.868 \text{ kip/ft} \approx \mathbf{850 \text{ lbs/ft}}$$

Correct answer is option C.

38. Problem

Correct Answer: (B) 4.5 ft X 4.5 ft

Explanation:

Ref to NCEES handbook page 3 for conversion factors.

$$1 \text{ ft} = 0.3048 \text{ m}$$

$$15 \text{ ft} = 0.3048 * 15 = 4.57 \text{ m}$$

For 4.57m , the allowable load in a 550 prop from the table = 20 kN

$$1 \text{ N} = 0.225 \text{ lbf}$$

$$1 \text{ kN} = 1000 * 0.225 \text{ lbf} = 0.225 \text{ kip}$$

$$20 \text{ kN} = 20 * 0.225 \text{ kip} = 4.5 \text{ kip}$$

$$\text{Total load on shoring posts} = 100 \text{ psf} + 75 \text{ psf} + 15 \text{ psf} = 190 \text{ psf} = 0.19 \text{ ksf}$$

$$\text{Allowable area} = \text{Allowable prop load} / \text{Load acting}$$

$$= 4.5 \text{ kip} / 0.19 \text{ ksf}$$

$$= 23.684 \text{ ft}^2$$

$$\text{Spacing} = \sqrt{\text{Area}}$$

$$= \sqrt{23.684 \text{ ft}^2}$$

$$= 4.866 \text{ ft}$$

Hence maximum spacing of 4.5 ftX 4.5 ft is permitted and Option B is the answer.

39. Problem

Correct Answer: (A) 50 lbs*ft

Explanation:

Per OSHA, CFR Title 29, Part 1926 ,Subpart L, section 1926.451(g)(4)(vii) and (ix),

Maximum load acting on mid rail for a guardrail with 100 pound top rail capacity i.e. single point adjustable suspension scaffold, = 75 lbs

Since the rail is welded between two posts, it can be assumed a fixed-fixed beam.

Design moment for simply supported beam (Ref NCEES PE handbook page 242 Case 16)

$$= Pl/8 = 75 \text{ lbs} * 5 \text{ ft} / 8$$

$$= 46.875 \text{ lbs} * \text{ft} \approx \mathbf{50 \text{ lbs} * \text{ft}}$$
 Correct answer is option A.

40. Problem

Correct Answer: (C)

Explanation:

Improving soil strength helps reduce the settlements.

Pile foundations transfer loads to a deeper depth away from the stress overlap zone between the two buildings and is also a remedial measure.

Placing lateral load resisting systems away from the perimeter of the building towards the center helps reduce the bearing pressures at overlap zones in the boundaries of adjacent buildings and helps reduce settlement.

Option C, placing the foundation of the new building above the foundation of the existing building does not help with the stress overlap and is the answer.

41. Problem

Correct Answer: (D) A-7-6(22)

Explanation:

Ref to NCEES PE handbook page 118, section 3.7.3 for AASHTO classification.

% of particles passing through No. 200 (0.075 mm sieve)

$$F = (1 - 0.35) * 100\%$$

$$F = 65\%$$

Hence it is between A-4 and A-7.

From page 111,

$$PI = LL - PL = 55 - 17 = 38$$

Liquid Limit = 51 i.e. greater than 41, and PI = 38 i.e. greater than 11

From page 119, AASHTO plasticity chart, it belongs to group A-7-6

Calculate group index from page 119

$$GI = (F - 35) [0.2 + 0.005(LL - 40)] + 0.01(F - 15)(PI - 10)$$

$$GI = (65 - 35) [0.2 + 0.005(55 - 40)] + 0.01(65 - 15)(38 - 10)$$

$$GI = 22.25, \text{ rounded off to } 22$$

Thus the correct answer is Option D, **A-7-6(22)**.

42. Problem

Correct Answer: (B) 24750 m³

Explanation:

Ref to NCEES PE handbook page 134 for volume weight relationships

For filling, required dry unit weight

$$\gamma_d = 1800 \text{ kg/m}^3$$

Then, weight of solids for 1 m³ = 1800 kg

From borrow pit, 1800 kg of solids will be required per m³ fill

Weight of water in soil from borrow pit = water content in borrow pit * weight of solids

$$= 10\% * 1800 \text{ kg} = 180 \text{ kg}$$

Total weight of soil required from borrow pit per m³ fill =

(Weight of solids + weight of water)

$$(1800 + 180) \text{ kg} = 1980 \text{ kg}$$

Given, density of borrow pit = 1600 kg/m³

For Weight of soil 1980 kg,

Volume required = Total Weight required / in situ density of borrow pit

$$= 1980 \text{ kg} / (1600 \text{ kg/m}^3)$$

$$= 1.2375 \text{ m}^3$$

Thus for every m³ of fill, 1.2375 m³ of soil from borrow pit has to be excavated

For 20000 m³ of fill, 1.2375 * 20,000 m³ = **24750 m³** has to be excavated.

Option B is the correct answer.

43. Problem

Correct Answer: (D)

Explanation:

Ref to ACI 318-14, page 318 Table 19.3.1.1

For sea water exposure, the exposure class is S1

From page 323, Table 19.3.2.1, for Class S1, the cementitious material specified is Type II.

Generally, chemical accelerators are not good to improve sulphate resistance.

Superplasticizers help lower the water cement ratio and help reduce the permeability of concrete. They help make concrete more durable and more resistant to sulphate attacks.

Hence Option D “Type II cement, sand, aggregates, water, super plasticizer” is the correct answer.

44. Problem

Correct Answer: (C) 60 bags

Explanation:

The weight ratio is 1 cement : 2 Fine aggregate : 3 Coarse aggregate

Calculate volume of each component per bag of cement i.e. per 94 lbf weight.

Cement Volume

$$= 1 * 94 \text{ lbf} / 195 \text{ lbf} / \text{ft}^3$$

$$= 0.482 \text{ ft}^3$$

Fine aggregate volume

$$= 2 * 94 \text{ lbf} / 160 \text{ lbf} / \text{ft}^3$$

$$= 1.175 \text{ ft}^3$$

Coarse aggregate volume

$$= 3 * 94 \text{ lbf} / 164 \text{ lbf} / \text{ft}^3$$

$$= 1.72 \text{ ft}^3$$

Water volume (Ref to NCEES PE handbook page 3 for volume of water per gallon)

$$= 6 \text{ gal} * 0.134 \text{ ft}^3 / \text{gal}.$$

$$= 0.804 \text{ ft}^3$$

Total volume yielded by 1 bag of cement

$$= (0.482 + 1.175 + 1.75 + 0.804) \text{ ft}^3$$

$$= 4.181 \text{ ft}^3$$

Number of bags required for 250 ft³

$$= 250 \text{ ft}^3 / 4.181 \text{ ft}^3$$

$$= 59.79$$

Hence **60 bags** are required and Option C is the correct answer

45. Problem

Correct Answer: (B) 18in

Explanation:

Diameter of #8 bar = 8/8 inch = 1 inch

Ref to ACI 318-14 page 427, section 25.4.9.2

Taking $\Psi_r = 1$

For normal weight concrete $\lambda = 1$

L_{dc} is the greater of

- a) $f_y \Psi_r / (50 \lambda \sqrt{f'_c}) \cdot d_b$
 $= 60000 \cdot 1 / (50 \cdot 1 \cdot \sqrt{6000}) \cdot 1 \text{ in} = 15.5 \text{ in}$
- b) $0.0003 \cdot f_y \Psi_r \cdot d_b$
 $= 0.0003 \cdot 60000 \cdot 1 \text{ in} = \mathbf{18 \text{ in}}$

Correct answer is option B.

46. Problem

Correct Answer: (D)

Explanation:

Option A is false because increase in temperature decreases the yield and ultimate strength of steel.

Option B is false because ductility of steel reduces with reducing temperature and the steel becomes more brittle.

Option C is false because the modulus of elasticity of steel increases with decreasing temperature.

Option D is true because decreasing temperatures result in increasing Modulus of Elasticity and consequently increase the stiffness of the members. Hence Option D is the answer.

47. Problem

Correct Answer: (B) I - A572 II- A500 III - A36

Explanation:

Per AISC Steel manual 15th edition, page 2-48, Table 2-4

For L sections (III) , Preferred specification is A36

For rectangular HSS section (II). Preferred specification is A500 Gr C

Section I is plate girder with plate thickness 1.5 inch and 2 inch

From page 2-50, Table 2-5 , preferred specification for plate is A572 or A36

Thus the correct answer is option B.

48. Problem

Correct Answer: (A) 11.6 ksi

Explanation:

Total depth of section = 15 in + 1 in (plate)

H = 16 in

Depth of extreme fiber of cover plate from the centroid

$y = H - 9.8 \text{ in}$

$y = 6.2 \text{ in}$

Section modulus = I/y

$= 2238 \text{ in}^4 / 6.2 \text{ in}$

$= 360.97 \text{ in}^3$

Flexural stress in cover plate = Moment /section modulus

$$= 350 \text{ kip} \cdot \text{ft} \cdot (12 \text{ in/ft}) / 360.97 \text{ in}^3$$

$$= \mathbf{11.6 \text{ ksi}}$$

Correct answer is Option A.

49. Problem

Correct Answer: (A) 794 ksi

Explanation:

Ref to NDS 2018 Supplement page 67 Table 5B

For 5 DF L1,

$E_{min} = 1060000 \text{ psi}$ (All adjustment values below are for E_{min})

From NDS 2018 page 11, Table 2.3.3, For 120°F , $C_t = 0.9$

From NDS supplement, page 66, $C_m = 0.833$

From NDS 2018 page 37 Table 5.3.1,

$$E'_{min} = E_{min} \cdot C_m \cdot C_t$$

$$E'_{min} = 1060000 \text{ psi} \cdot 0.833 \cdot 0.9$$

$$E'_{min} = 794682 \text{ psi}$$

Hence the correct answer is option A, **794 ksi**.

50. Problem

Correct Answer: (B)

Explanation:

Option A is correct per TMS 402/602-16, page S-30 to S-33, section 1.8C and 1.8D.

Options C and D are correct per TMS 402/602-16, page S-36, Table SC-1 and SC-2,

If the mortar is stronger than the individual masonry units, the masonry units will crack before the mortar which is not the desired mode of failure of the overall masonry system.

Hence it is always advisable for the mortar to be weaker than the units. Option B is false and the answer.

51. Problem

Correct Answer: (A)

Explanation:

Samples of the concrete are made at the point where the mix is delivered i.e. the point of placement. Hence, Option B is correct

Per ACI 318-14, page 480, Section 23.12.3.1 (a), Options C and Option D are correct.

Per ACI 318-14, page 478 section 26.12.1.1 (a), an average of three 4X8 cylinders are required. Hence Option A is not true and is the answer.

52. Problem

Correct Answer: (B) 1750 psi

Explanation:

Ref to Unit strength method from TMS 402/602, Masonry specification page S-18, Table-2.

If individual masonry unit of 2000 psi strength is used with Type N mortar, the specified strength of masonry assembly is given as

Net area compressive strength of concrete masonry, psi (MPa) ¹	Net area compressive strength of ASTM C90 concrete masonry units, psi (MPa)	
	Type M or S mortar	Type N mortar
1,750 (12.07)	---	2,000 (13.79)

Correct answer is Option B, 1750 psi.

53. Problem

Correct Answer: (B) W18X50

Explanation:

For compact sections, bending about minor axis, lateral torsional buckling and flange local buckling don't apply.

From AISC steel manual, Part ASIC 360-16, page 16.1-56, section 2(a), W18 sections are compact, hence only yielding controls.

Maximum moment by uniformly distributed load =

$$w \cdot L^2 / 8$$

$$M = 0.5 \text{ kip/ft} \cdot (25 \text{ ft})^2 / 8 = 39.06 \text{ kip} \cdot \text{ft}$$

W18×50	16.6	41.4
W16×50	16.3	40.7
W8×35	16.1	40.2
W24×62	15.7	39.1
W21×48 ^f	14.9	36.7
W21×57	14.8	36.9
W16×45	14.5	36.2
W8×31 ^f	14.1	35.1
W10×22	14.0	34.9

From steel manual page 3-32, Table 3-4

Lightest W18 section that satisfies moment M_{ny}/Ω is

W18X50 with $M_{ny}/\Omega = 41.4 \text{ kip}\cdot\text{ft}$

Correct answer is Option B

54. Problem

Correct Answer: (C) 234 kips

Explanation:

Ref to AISC steel manual 15th ed., page 1-38, Table 1-5

For C15X40

$d = 15 \text{ in}$

$t_w = 0.52 \text{ in}$

$h = T = 12.125 \text{ in}$

$E = 29000 \text{ ksi}$

$F_y = 50 \text{ ksi}$

From AISC steel manual, Page 16.1-71, (b)(2)

$k_v = 5.34$ for webs without stiffeners

$1.1 \sqrt{(k_v \cdot E / F_y)}$

$= 1.1 \sqrt{(5.34 \cdot 29000 / 50)}$

$= 61.2$

$h/t_w = 12.125 \text{ in} / 0.52 \text{ in} = 23.3$

Thus $h/t_w < 1.1 \sqrt{(k_v \cdot E / F_y)}$

From Page 16.1-71, (b)(1) $C_v = 1$

From Page 16.1-70, equation G2-1

$V_n = 0.6 \cdot F_y \cdot A_w \cdot C_v$

$V_n = 0.6 \cdot 50 \text{ ksi} \cdot 15 \text{ in} \cdot 0.52 \text{ in} \cdot 1$

$V_n = 0.6 \cdot 50 \text{ ksi} \cdot 15 \text{ in} \cdot 0.52 \text{ in} \cdot 1$

$V_n = 234 \text{ kips}$. Hence the closest answer is Option C, **234 kips**.

55. Problem

Correct Answer: (D) 27.5 psi

Explanation:

Shear depth, $d_v = 24$ in

Total shear stress $F_v = V / (b * d_v)$

$F_v = 13000 \text{ lbs} / (7.625 \text{ in} * 24 \text{ in}) = 71.04 \text{ psi}$

Calculate F_{vm} , shear stress carried by masonry

From TMS 402/602-16, page C-118, eq 8-26,

$F_{vm} = 0.5 * [\{ 4.0 - 1.75 * (M / V d_v) \} * \sqrt{f'_m}] + 0.25 * P / A_n$ ($P = 0$ here)

$F_{vm} = 0.5 * [\{ 4.0 - 1.75 * 1 \} * \sqrt{1500}] = 43.57 \text{ psi}$

From eq. 8-22, for F_{vs} i.e. shear carried by reinforcement,

$F_v = (F_{vm} + F_{vs}) * \gamma_g$ ($\gamma_g = 1.0$ for fully grouted section)

$F_{vs} = 71.04 \text{ psi} - 43.57 \text{ psi} \approx \mathbf{27.5 \text{ psi}}$

Correct answer is Option D.

56. Problem

Correct Answer: (B) 1

Explanation:

24F-V1 SP/SP denotes that the timber is of Southern Pine species.

From NDS-2018 -Supplement page 62,

$x = 20$ for southern pine

$L = 15\text{ft}$ for simply supported beam

b = 4 in

d = 16 in

$$C_v = (21/l)^{\frac{1}{x}} (12/d)^{\frac{1}{x}} (5.125/b)^{\frac{1}{x}}$$

$$C_v = (21/15)^{\frac{1}{20}} (12/16)^{\frac{1}{20}} (5.125/4)^{\frac{1}{20}}$$

$C_v = 1.01$, must be less than or equal to 1

Hence Option B, **1** is the correct answer.

57. Problem

Correct Answer: (D) 10in

Explanation:

Ref to ACI 318-14, page 84, Table 7.3.1.1

For one end continuous i.e. exterior span,

$$t = L/24 = 17 \text{ ft} / 24 = 8.5 \text{ in}$$

For both ends continuous i.e. interior span,

$$t = L/28 = 21 \text{ ft} / 28 = 9 \text{ in (governs)}$$

From page 84, section 7.3.1.1.2, for unit weight 110 pcf,

$$\text{Modification factor} = 1.65 - 0.005 \cdot w_c$$

$$= 1.65 - 0.005 \cdot 110 = 1.1$$

$$\text{Required thickness} = 9 \text{ in} \cdot 1.1 = 9.9 \text{ in}$$

Hence the correct answer is Option D, **10in** .

58. Problem

Correct Answer: (B) 9 in

Explanation:

Here, diameter of stirrups to be used is not given, however, ACI 318-14 limits the tie spacing regardless of tie diameter.

From ACI 318-14, page 341 Table 21.2.1, strength reduction factor for shear

$$\Phi = 0.75$$

Effective depth of section

$d = 24 \text{ in} - \text{cover}$

$d = 24 \text{ in} - 2.5 \text{ in} = 21.5 \text{ in}$

$b_w = 12 \text{ in}$

From ACI 318-14, page 316, Table 19.2.4.2, for normal weight concrete, $\lambda = 1$

From ACI 318-14, page 354, section 22.5.5.1,

Shear capacity provided by concrete

$$V_c = 2\lambda(\sqrt{f'_c}) * b_w * d$$

$$V_c = 2 * 1 * (\sqrt{4000}) * 12 \text{ in} * 21.5 \text{ in}$$

$$V_c = 32634.7 \text{ lbs} = 32.63 \text{ kips}$$

From ACI 318-14, page 359, section 22.5.10

$$V_s = V_u / \Phi - V_c$$

$$V_s = 70 \text{ kips} / 0.75 - 32.63 \text{ kips}$$

$$V_s = 60.7 \text{ kips}$$

Ref to ACI 318-14 page 146, Table 9.7.6.2.2 for max permissible spacing of stirrups

$$4(\sqrt{f'_c}) * b_w * d$$

$$= 4(\sqrt{4000}) * 12 \text{ in} * 21.5 \text{ in}$$

$$= 65269.4 \text{ lbs} = 65.27 \text{ kips}$$

$$\text{Here } V_s < 4(\sqrt{f'_c}) * b_w * d$$

Thus the maximum permissible spacing is the least of $d/2$ or 24 in

$$s = d/2$$

$$s = 21.5 \text{ in} / 2$$

$$s = 10.75 \text{ in}$$

Hence spacing cannot be greater than 10.75 in. Here, the correct answer is Option B, **9 in.**

59. Problem

Correct Answer: (C) 250 kips

Explanation:

$$K_{lx} = 28 \text{ ft} = L_{cx}$$

$$K_{ly} = 14 \text{ ft}$$

From AISC steel manual 15th ed., page 1-29, table 1-1

For W8X40, $r_x = 3.53 \text{ in}$, $r_y = 2.04 \text{ in}$

Since $K_{lx} > K_{ly}$ from steel manual page 4-5, equation 4-1

Equivalent effective length,

$$L_{cy,eq} = L_{cx} / (r_x/r_y)$$

$$= 28 \text{ ft} / (3.53 \text{ in} / 2.04 \text{ in})$$

= 16.18 ft > $K L_y$, so use this effective length.

From steel manual page 4-24, table 4-1 a

For W8X40 with $L_c = 17 \text{ ft}$, $\Phi P_n = 253 \text{ kips}$

Hence the correct answer is Option C, **250 kips**.

60. Problem

Correct Answer: (B) I and II

Explanation:

From AISC steel manual page 15th ed. Page 16.1-38, section E5

Effects of eccentricity can be ignored if

L_c/r as determined in the section does not exceed 200

Members are loaded at the ends in compression through the same one leg.

Condition III is not a requirement.

Hence, the correct answer is Option B, **I and II**.

61. Problem

Correct Answer: (C) 233 kips

Explanation:

From AISC Steel Manual 15th ed. page 4-17, table 4-1a

For W14X82 with 15ft effective length

Allowable compression, $P_c = 489 \text{ kips}$

Lateral torsional buckling is not a concern so get allowable yielding moments.

From steel manual page 3-25, table 3-2 for W14X82, $M_{cx} = M_{px}/\Omega = 347 \text{ kip}\cdot\text{ft}$

From steel manual page 3-31, table 3-4 for W14X82, $M_{cy} = M_{ny}/\Omega = 112 \text{ kip}\cdot\text{ft}$

From steel manual page 16.1-77, eq H1-1a for interaction,

$$P_r/P_c + (8/9) * (M_{rx}/M_{cx} + M_{ry}/M_{cy}) = 1$$

$$P_r/489 \text{ kips} + (8/9) * (80 \text{ kip}\cdot\text{ft} / 347 \text{ kip}\cdot\text{ft} + 40 \text{ kip}\cdot\text{ft} / 112 \text{ kip}\cdot\text{ft}) = 1$$

$$Pr/489 \text{ kip} = 0.478$$

$$Pr = 233.7 \text{ kips}$$

Correct answer is option C, **233 kips**.

62. Problem

Correct Answer: (B) 5 kips

Explanation:

From NDS-2018 supplement page 14, Table 1B, standard dressed size for 2X6 = 1.5X5.5

From NDS-2018 supplement page 34, Table 4A, for select structural DFL,

Compression parallel to grain $F_c = 1700 \text{ psi}$

From NDS-2018, page 29, Table 4.3.1

$$F_c' = F_c * C_d * C_m * C_t * C_f * C_i * C_p$$

$$F_c' = 1700 \text{ psi} * 1 * 1 * 1 * 1.15 * 1 * 0.33$$

$$F_c' = 645.15 \text{ psi}$$

Allowable compression, $P = F_c' * A$

$$P = 645.15 \text{ psi} * 1.5 \text{ in} * 5.5 \text{ in}$$

$$P = 5322.5 \text{ lbs}$$

Correct answer is Option B, **5 kips**.

63. Problem

Correct Answer: (B) 38.3 kips

Explanation:

Here, information on ties is not provided. Hence, only the strength of concrete is calculated.

Effective cover for columns is assumed as 3 in.

$$d = 20 \text{ in} - 3 \text{ in}$$

$$d = 17 \text{ in}$$

$$b_w = 20 \text{ in}$$

$$\lambda = 1 \text{ (Assume normal weight concrete)}$$

$$A_g = 20 \text{ in} * 20 \text{ in} = 400 \text{ in}^2$$

$$f'_c = 4000 \text{ psi}$$

From ACI 318-14, page 354, equation 22.5.6.1

$$V_c = 2 \left[1 + \frac{N_u}{(2000 * A_g)} \right] * \lambda * (\sqrt{f'_c}) * b_w * d$$

$$V_c = 2 \left[1 + \frac{150000 \text{ lbs}}{(2000 * 400 \text{ in}^2)} \right] * 1 * (\sqrt{4000}) * 20 \text{ in} * 17 \text{ in}$$

$$V_c = 51070.78 \text{ lbs}$$

From ACI 318-14, page 341, Table 21.2.1

For shear, $\Phi = 0.75$

$$\text{Design strength} = \Phi * V_c = 0.75 * 51070.78 \text{ lbs} \approx \mathbf{38.3 \text{ kips}}$$

Correct answer is Option B.

64. Problem

Correct Answer: (D) Combined moment frame and shear wall

Explanation:

Cantilevered columns provide the least redundancy.

Flat slabs with drop panels primarily resist gravity loads and are not for lateral application.

Reinforced masonry walls have average seismic performance. Dual systems i.e. combined wall and moment frame systems have the highest performance and redundancy. Hence Option D is the answer.

65. Problem

Correct Answer: (C) 98

Explanation:

From AISC Steel manual 15th edition, page 1-22, Table 1-1,

$$\text{Area of W18X50, } A_s = 14.7 \text{ in}^2$$

Ref to AISC steel manual 15th edition, page 16.1-96, section 2d.1

For full composite action, the nominal shear force between the steel beam and the concrete slab transferred by steel anchors V' , between the point of maximum positive moment and the point of zero moment shall be determined as the lowest value in accordance with the limit states of

- a) concrete crushing, (equation I3-1a, page 16.1-97)

$$V' = 0.85 * f'_c * A_c$$

$$V' = 0.85 * 4 \text{ ksi} * (6 \text{ ft} * 5 \text{ in}) = 1224 \text{ kips}$$

- b) tensile yielding of the steel section (equation I3-1b, page 16.1-97)

$$V' = F_y \cdot A_s$$

$$V' = 50 \text{ ksi} \cdot 14.7 \text{ in}^2 = 735 \text{ kips (governs)}$$

The point of max moment to zero moment in a simply supported beam with Uniformly distributed load is half the span length

$$\text{Number of studs required for half the span} = V' / Q_n = 735 \text{ kips} / 15 \text{ kips} = 49$$

$$\text{Number of studs required for full span} = 49 \cdot 2 = \mathbf{98}$$

Correct answer is option C

66. Problem

Correct Answer: (C) 63.8 kips

Explanation:

Area of stud

$$A_s = \pi \cdot \Phi^2 / 4 = \pi \cdot (1 \text{ in})^2 / 4 = 0.785 \text{ in}^2$$

$$F'_c = 6000 \text{ psi} = 6 \text{ ksi}$$

$$E = 4400 \text{ ksi}$$

From AISC steel manual 15th ed., page 16.1-105, eqn. I8-1

$$Q_n = 0.5 \cdot A_s \cdot \sqrt{f'_c \cdot E}$$

$$Q_n = 0.5 \cdot 0.785 \text{ in}^2 \cdot \sqrt{6 \text{ ksi} \cdot 4400 \text{ ksi}}$$

$$Q_n = 0.5 \cdot 0.785 \text{ in}^2 \cdot \sqrt{6 \text{ ksi} \cdot 4400 \text{ ksi}}$$

$$Q_n = \mathbf{63.8 \text{ kips}}$$

Correct answer is Option C.

67. Problem

Correct Answer: (B) W12X40

Explanation:

$$\text{Linear load due to weight of 13 ft tall wall} = 80 \text{ lbf/ft}^2 \cdot 13 \text{ ft} = 1.04 \text{ kip/ft}$$

$$\text{Linear load due to floor dead load} = 75 \text{ lbf/ft}^2 \cdot (8 \text{ ft} / 2) = 0.3 \text{ kip/ft}$$

$$\text{Linear load due to floor live load} = 100 \text{ lbf/ft}^2 \cdot (8 \text{ ft} / 2) = 0.4 \text{ kip/ft}$$

$$\text{Total load} = 1.04 \text{ kip/ft} + 0.3 \text{ kip/ft} + 0.4 \text{ kip/ft} = 1.74 \text{ kip/ft}$$

$$\text{Moment due to uniform loading on a simply supported beam} = w \cdot L^2 / 8$$

$$M = 1.74 \text{ kip/ft} \cdot (25 \text{ ft})^2 / 8 = 135.93 \text{ kip}\cdot\text{ft}$$

$$\text{Maximum shear} = w \cdot L / 2$$

$$V = 1.74 \text{ kip/ft} * 25 \text{ ft} / 2 = 21.75 \text{ kip}$$

Since compression flanges are fully supported, lateral torsional buckling won't control.

From AISC steel manual 15th edition, page 3-26, Table 3-2,

All beams satisfy shear requirements.

Allowable plastic moment capacity for W12X30 = 108 kip*ft, not enough.

Allowable plastic moment capacity for **W12X40** = 142 kip*ft, O.K.

Hence, the correct answer is Option B.

68. Problem

Correct Answer: (A) 15/32 in

Explanation:

From AISC steel manual 15th ed. , page 16.1-130, Table J3.3, nominal hole size for standard holes with 1/2 in. diameter bolts = 9/16 in

From AISC steel manual 15th ed. , page 16.1-132, Table J3.4, minimum edge distance from center of hole to the edge of connected part for 1/2 in. diameter bolts = 3/4 in

Clear distance between the edge of plate and the hole

$$= 3/4 \text{ in} - (9/16 \text{ in})/2$$

$$= \mathbf{15/32 \text{ in}}$$

Correct answer is Option A.

69. Problem

Correct Answer: (C) 26 kips

Explanation:

From AISC steel manual 15th ed. , page 16.1-130, Table J3.3, nominal hole size for standard holes with 3/4 in. diameter bolts = 13/16 in

For edge bolt,

$$\text{Clear distance, } l_c = 2 \text{ in} - (13/16 \text{ in})/2 = 1.594 \text{ in}$$

For interior bolts,

$$\text{Clear distance, } l_c = 2.5 \text{ in} - (13/16) \text{ in} = 1.688 \text{ in}$$

Hence edge bolt governs

From AISC steel manual 15th ed. , page 16.1-136, eqn. J3-6a

$$\text{Nominal Bearing strength} = 2.4 * d * t * F_u$$

$$= 2.4 * 0.75 \text{ in} * 0.5 \text{ in} * 58 \text{ ksi} = 52.2 \text{ kips}$$

From AISC steel manual 15th ed. , page 16.1-136, eqn. J3-6c

Nominal Tearout strength = $1.2 * l_c * t * F_u$

$$= 1.2 * 1.594 \text{ in} * 0.5 \text{ in} * 58 \text{ ksi} = 55.5 \text{ kips}$$

Allowable bearing/tearout strength = Nominal strength / Ω

$$= 52.2 \text{ kips} / 2 \approx \mathbf{26 \text{ kips}}$$

Hence the correct answer is Option C.

70. Problem

Correct Answer: (B) 5/8 in

Explanation:

Ref to AISC steel manual 15th ed. Page 9-4, Fig 9-1 for width of Whitmore section.

Distance between center of end bolts

$$L = 3 * 2.5 \text{ in} = 7.5 \text{ in}$$

Effective width of gusset

$$L_w = 2 * L * \tan (30^\circ)$$

$$L_w = 2 * 7.5 \text{ in} * \tan (30^\circ) = 8.66 \text{ in}$$

Gross area of plate = 8.66 * thickness

$$A_g = 8.66 \text{ in} * t$$

From AISC steel manual 15th ed.

Page 16.1-137, eqn. J4-1 ($\Omega = 1.67$)

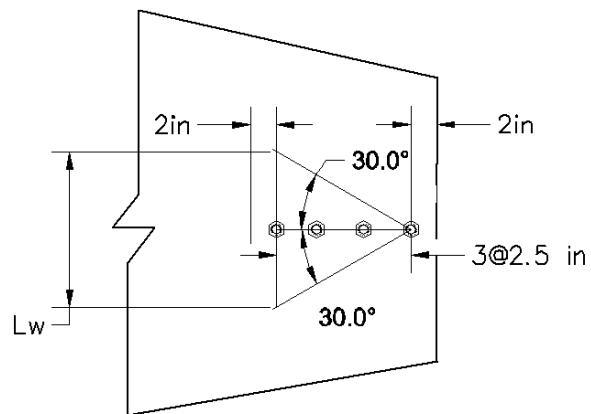
For yielding, Allowable strength = Service load = 100 kips

$$\text{Allowable strength} = F_y * A_g / \Omega$$

$$100 \text{ kips} = 36 \text{ ksi} * 8.66 \text{ in} * t / 1.67$$

$$t = 100 \text{ kips} * 1.67 / (36 \text{ ksi} * 8.66 \text{ in}) = 0.54 \text{ in}$$

Hence minimum thickness from the option is **5/8 in** . Correct answer is Option B.



71. Problem

Correct Answer: (D)

Explanation:

The grade of steel does not affect the rotational stiffness, only the strength of the joint.

The beam web welds primarily resist shear.

The diameter of the bolts used in the flange plates only increases the capacity of the connection. The geometry and thickness of the flange plate and angles primarily affect the rotational stiffness. Correct answer is Option D.

72. Problem

Correct Answer: (D) 12.5 kips

Explanation:

The tipping point is the tip of the horizontal leg of the angle.

Overturning moment = 5 kips * 5 in

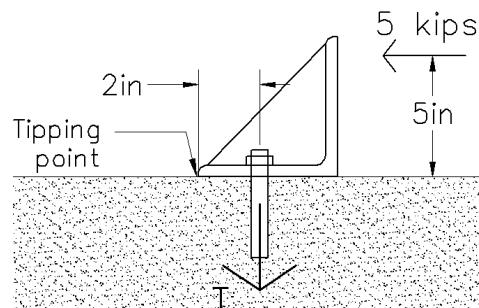
Resisting moment = $T * 2$ in

$T * 2$ in = 5 kips * 5 in

$T = 5 \text{ kips} * 5 \text{ in} / 2 \text{ in}$

$T = 12.5 \text{ kips}$

Correct answer is Option D.



73. Problem

Correct Answer: (C) 62.4 kips

Explanation:

From AISC steel manual 15th edition, page 8-68, Table 8-4,

$e_x = a * l$,

i.e. $a = 3 \text{ in} / 6 \text{ in} = 0.5$

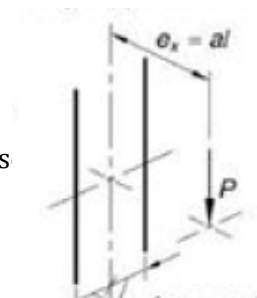
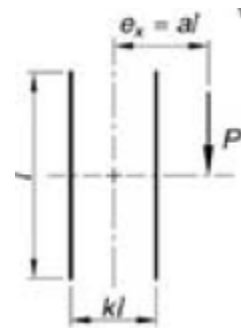
$K * l = 6 \text{ in}$

$k = 6 \text{ in} / 6 \text{ in} = 1$

For $a = 0.5$ and $k = 1$, $C = 2.6$

Size of 1/2 inch weld in 16th of an inch,

1/2 in = 8/16 in. Hence D = 8



From AISC steel manual 15th edition, page 8-67, Table 8-3,

$C_1 = 1$ for $F_{exx} = 70$ ksi

Allowable Load = $C * C_1 * D * 1 / \Omega$

$$= 2.6 * 1 * 8 * 6 / 2$$

$$= \mathbf{62.4 \text{ kips}}$$

Correct answer is Option C.

74. Problem

Correct Answer: (A) 1

Explanation:

The slab is not restrained on all four sides and not a part of seismic force resisting system.

From ACI 318-14 page 507, Appendix A, area of #4 tie = 0.2 in^2

Effective area of tie in transverse shear, $A = 2 * 0.2 \text{ in}^2 = 0.4 \text{ in}^2$

From ACI 318-14, page 204, section 15.4.2, tie spacing shall be at least the greater of

$$\text{a) } A = 0.75 * (\sqrt{f'_c}) * b * s / f_y$$

$$s = A * f_y / [0.75 * (\sqrt{f'_c}) * b] = 0.4 \text{ in}^2 * 60000 / [0.75 * (\sqrt{4000}) * 18 \text{ in}] = 28.1 \text{ in}$$

$$\text{b) } A = 50 * b * s / f_y$$

$$s = A * f_y / [50 * b] = 0.4 \text{ in}^2 * 60000 / [50 * 18 \text{ in}] = 26.6 \text{ in}$$

Since the tie spacing is greater than the joint depth available i.e. 8 in, only **1** tie is required.

Correct answer is Option A.

75. Problem

Correct Answer: (C) 1826 lbs

Explanation:

From NDS 2018 supplement page 14, Table 1B

Dressed size of 6X8 = 5.5 in X 7.5 in (main member)

Dressed size of 4X6 = 3.5 in X 5.5 in (side member)

From NDS 2018, page 94, Table 12A,

For DFL with $t_m = 5.5$ in and $t_s = 3.5$ in

$$Z_{m \perp} = 1390 \text{ lbs} \quad Z_{||} = 2660 \text{ lbs} \quad (\text{no adjustment required since all values} = 1)$$

From NDS 2018, page 176, Appendix J, equation J-3

$$Z' = (Z_{||} * Z_{m \perp}) / (Z_{||} * \sin^2 \Theta + Z_{||} * \cos^2 \Theta)$$

$$Z' = (2660 * 1390) / (2660 * \sin^2 45^\circ + 1390 * \cos^2 45^\circ)$$

$$Z' \approx \mathbf{1826 \text{ lbs}}$$

Correct answer is Option C.

76. Problem

Correct Answer: (A)

Explanation:

Option B, C and D are all true about controlling differential settlement in buildings.

Option A is false. Trees and vegetation absorb moisture from soil and cause shrinkage and uneven settlement in the surrounding soil. Roots of trees around foundations can also exert pressure that leads to foundation movement and settlement. Hence, Option A is the answer.

77. Problem

Correct Answer: (B) 22 ft

Explanation:

For uniform bearing pressure, the resultant of the total load must lie at the center of the combined footing.

Total resultant Vertical load, $P = 50 \text{ kips} + 40 \text{ kips} + 100 \text{ kips} + 80 \text{ kips} = 270 \text{ kips}$

Location of the resultant from left edge = $L/2$

Taking first moment about the left edge,

$$P * L/2 = \text{Right column load} * (15\text{ft} + 1\text{ft}) + \text{Left column load} * 1\text{ft}$$

$$270 \text{ kip} * L/2 = (100 \text{ kip} + 80 \text{ kip}) * 16 \text{ ft} + (50 \text{ kip} + 40 \text{ kip}) * 1 \text{ ft}$$

$$L = 2 * 2970 \text{ kip*ft} / 270 \text{ kip*ft} = \mathbf{22 \text{ ft}}$$

Hence length of the footing must be 22ft for uniform bearing pressure.

Correct answer is Option B.

78. Problem

Correct Answer: (C) 622 kips

Explanation:

From ACI 318-14, page 362, section 22.6.4.1,

Critical perimeter, $b_o = 4 \cdot (21\text{in}/2 + 18\text{in} + 21\text{in}/2) = 4 \cdot 39\text{ in} = 156\text{ in}$

Column at the center of footing is considered 'interior'

From ACI 318-14, page 365, section 22.6.5.3, $\alpha = 40$

From ACI 318-14, page 364, Table 22.6.5.2, $\beta = 18\text{in}/18\text{in} = 1$

v_c is the least of

a) $4 \cdot (\sqrt{f'_c}) = 4 \cdot 1 \cdot \sqrt{4000} = 252.98\text{ psi}$

b) $(2 + 4/\beta) \cdot \sqrt{f'_c} = (2 + 4/1) \cdot 1 \cdot \sqrt{4000} = 379.47\text{ psi}$

c) $(2 + \alpha \cdot d/b_o) \cdot \sqrt{f'_c} = (2 + 40 \cdot 21\text{ in}/156\text{ in}) \cdot 1 \cdot \sqrt{4000} = 467.04\text{ psi}$

$v_c = 252.98\text{ psi}$

$V_n = v_c \cdot b_o \cdot d = 252.98\text{ psi} \cdot 156\text{ in} \cdot 21\text{ in} = 828762.5\text{ lbs} = 828.8\text{ kips}$

From ACI 318-14, page 341, Table 21.2.1, $\Phi = 0.75$ for shear

Design strength $= \Phi \cdot V_n = 0.75 \cdot 828.8\text{ kips} \approx \mathbf{622\text{ kips}}$

Correct answer is Option C.

79. Problem

Correct Answer: (B) 2.5 ft

Explanation:

Thrust, $C = 75\text{ kips}$

Uplift, $U = 45\text{ kips}$

Check required diameter for uplift

Uplift = Allowable skin friction resistance

$$45 \text{ kips} = L * \pi * D * 300 \text{ psf}$$

$$D = 45 \text{ kips} / (20 \text{ ft} * \pi * 0.3 \text{ ksf})$$

$$D = 2.387 \text{ ft}$$

Check this diameter for thrust capacity

Thrust $\leq$ Allowable skin friction resistance + Allowable end bearing resistance

$$75 \text{ kips} \leq L * \pi * D * 0.3 \text{ ksf} + (\pi * D^2 / 4) * 12 \text{ ksf}$$

$$75 \text{ kips} \leq 98.7 \text{ kips O.K.}$$

Hence the correct answer is Option B, **2.5 ft.**

80. Problem

Correct Answer: (D)

Explanation:

The cantilevered retaining wall will be subjected to lateral earth pressure from left to right that will cause tension in the areas marked by red in the figure.

Those areas need tensile reinforcement.

The correct answer is Option D.

